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Protocol

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Web-based multi-omics integration using the Analyst software suite

Jessica D. Ewald[1], Guangyan Zhou[1], Yao Lu[2], Jelena Kolic[3], Cara Ellis[4], James D. Johnson[3], Patrick E. Macdonald[4] & Jianguo Xia[1,2]

Abstract

The growing number of multi-omics studies demands clear conceptual workflows coupled with easy-to-use software tools to facilitate data analysis and interpretation. This protocol covers three key components involved in multi-omics analysis, including single-omics data analysis, knowledge-driven integration using biological networks and data-driven integration through joint dimensionality reduction. Using the dataset from a recent multi-omics study of human pancreatic islet tissue and plasma samples, the first section introduces how to perform transcriptomics/proteomics data analysis using ExpressAnalyst and lipidomics data analysis using MetaboAnalyst. On the basis of significant features detected in these workflows, the second section demonstrates how to perform knowledge-driven integration using OmicsNet. The last section illustrates how to perform data-driven integration from the normalized omics data and metadata using OmicsAnalyst. The complete protocol can be executed in ~2 h. Compared with other available options for multi-omics integration, the Analyst software suite described in this protocol enables researchers to perform a wide range of omics data analysis tasks via a user-friendly web interface.

Key points

- This protocol for web-based multi-omics integration covers single-omics data analysis using ExpressAnalyst and MetaboAnalyst, followed by knowledge-driven integration using OmicsNet and data-driven integration using OmicsAnalyst.
- This series of web-based tools allows researchers to perform a wide range of omics data analysis tasks via a user-friendly web interface, helping to democratize omics data analysis and empower researchers without strong statistics and programming backgrounds.

Key references

Zhou, G. et al. *Nucleic Acids Res.* **50**, W527–W533 (2022): https://doi.org/10.1093/nar/gkac376

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[1]Institute of Parasitology, McGill University, Montreal, Quebec, Canada. [2]Department of Microbiology and Immunology, McGill University, Montreal, Quebec, Canada. [3]Life Sciences Institute, Department of Cellular and Physiological Sciences, University of British Columbia, Vancouver, British Columbia, Canada. [4]Department of Pharmacology, University of Alberta, Edmonton, Alberta, Canada. e-mail: jeff.xia@mcgill.ca

Introduction

Multi-omics data acquisition has become commonplace across life sciences, enabling comprehensive molecular characterizations of biology and disease[1]. Different analytical strategies have been proposed for disease diagnosis, drug discovery and systems biology[1–3]. However, corresponding data analysis workflows remain difficult to generalize, posing strong barriers for researchers who are interested in conducting these types of studies[4]. 'Multi-omics integration' is an umbrella term used to describe many distinct approaches developed for a diversity of experimental designs and study contexts[4,5]. In many cases, these different approaches have special assumptions or requirements in terms of omics type, sample size, location or temporal compatibility, or data preprocessing, to name just a few[3]. Compared with the practices of single-omics data analysis, which have largely converged into one or a few well-established workflows[6,7], the current landscape of multi-omics integration is highly fragmented, dominated by approaches featuring complex multivariate statistics, machine learning algorithms and custom-built knowledgebases[8,9]. Researchers are expected to have advanced statistics training, programming skills and familiarity with multiple bioinformatics resources to execute the diverse tasks involved in multi-omics data analyses[8,10].

Computational strategies for multi-omics integration can be broadly categorized into two main types: knowledge driven and data driven[11]. In knowledge-driven approaches, data from each omics layer is analyzed individually to identify molecular features of interest. These features, also known as seeds, are subsequently mapped to curated knowledgebases containing annotations in terms of ontologies, pathways or interactions for multiple types of molecules. The process will produce annotated and enriched molecules that can be intuitively visualized as trees, heat maps or networks to reveal emerging themes across omics layers[12,13]. Knowledge-driven approaches are conceptually simple and are largely agnostic to experimental designs. Their applications are primarily limited by the quality and coverage of the underlying knowledgebases and therefore are suitable for well-studied model organisms and well-characterized omics data types[14–16]. On the other hand, data-driven approaches perform de novo identification of shared patterns and correlations across different omics layers[17,18]. After key trends have been uncovered, the main contributing features are examined to infer their functional implications[19,20]. In general, data-driven approaches are not limited by our prior knowledge, allowing for identification of novel relationships[19]. However, larger sample sizes and strictly matching samples are required to perform data-driven integration, and extra steps are often necessary to reveal the molecule features underlying the identified patterns for data interpretation[21]. Whenever possible, knowledge-driven and data-driven approaches should be used together to achieve comprehensive data understanding.

Development of the protocol

To help democratize omics data analysis and to empower researchers without strong statistics and programming backgrounds, we have developed a series of web-based tools to assist researchers to perform a wide range of omics data analysis tasks via a user-friendly web interface. We have previously published several Nature Protocols on using our web-based tools for analysis of individual omics data, including MetaboAnalyst (www.metaboanalyst.ca) for targeted and untargeted metabolomics data[22,23], and MicrobiomeAnalyst for marker gene survey and shotgun metagenomics data[24], as well as ExpressAnalyst (www.expressanalyst.ca; previously called NetworkAnalyst) for transcriptomics and proteomics data[25]. These tools have been used to perform >2.5 million analysis jobs in the last 12 months. In addition, we have also developed miRNet for microRNA (miRNA) data analysis and mGWAS-Explorer for metabolite genome-wide association studies data[26,27]. Building upon this strong foundation of single-omics tools, we recently also developed OmicsNet (www.omicsnet.ca) and OmicsAnalyst (www.omicsanalyst.ca) to support knowledge-driven and data-driven multi-omics integration, respectively[16,28,29]. In this protocol, we provide a conceptual framework coupled with step-by-step instructions on how to leverage this suite of web-based tools for comprehensive multi-omics analyses.

Overview of the procedure

This protocol first illustrates the general workflow of single-omics analysis followed by knowledge-driven and data-driven multi-omics integration, respectively (Fig. 1). Before performing integration, each omics dataset should be first analyzed individually[20].

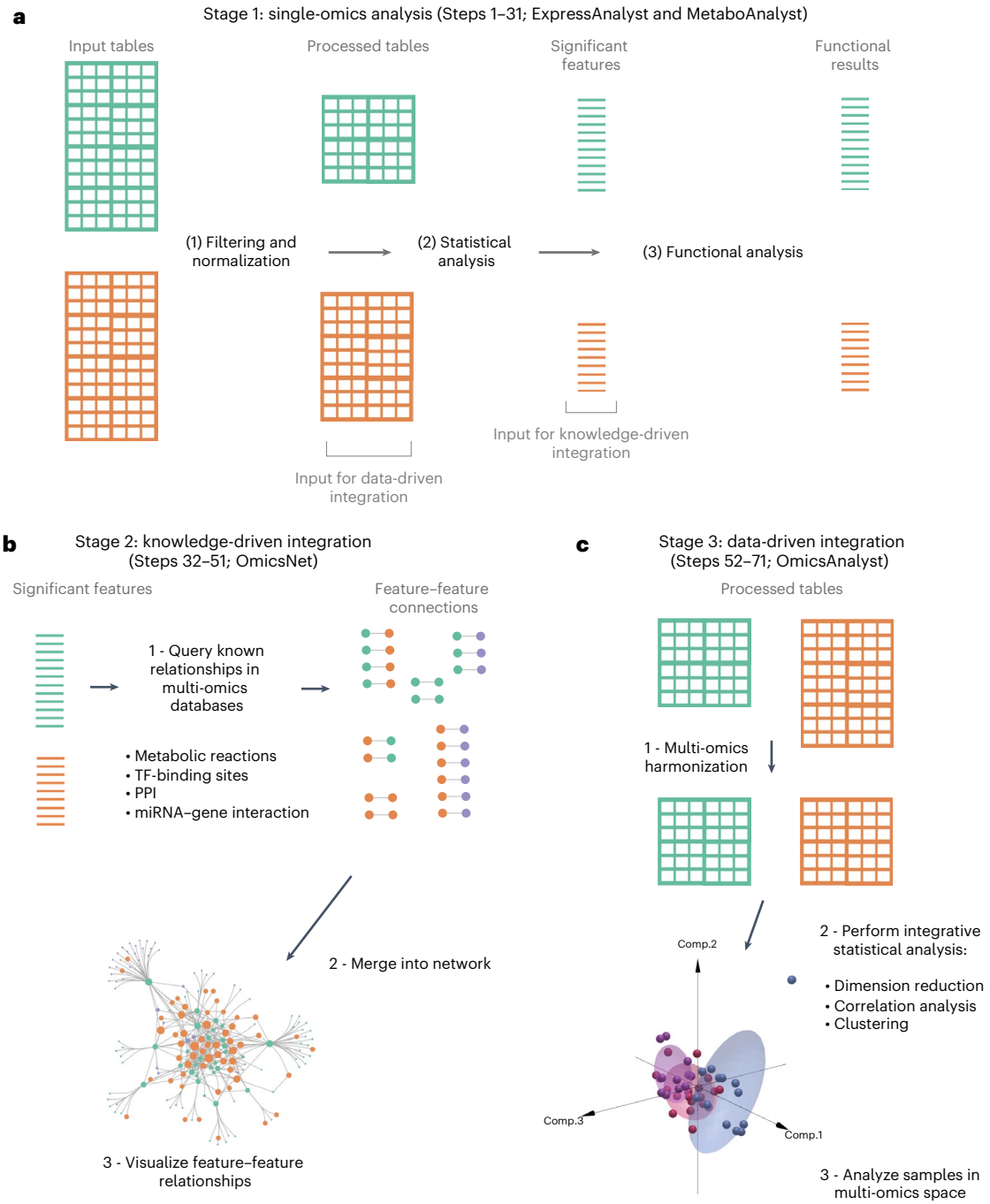


Fig. 1 | Overview of multi-omics analysis pipelines. a–**c**, The analysis of a multi-omics dataset should start with robust, single-omics analysis (**a**). The lists of significant features and normalized tables from the single-omics analysis can be used for knowledge-driven (**b**) and data-driven (**c**) integration. MetaboAnalyst, ExpressAnalyst and MicrobiomeAnalyst are used for (**a**) protocol Stage 1, which covers the proper filtering, normalization and statistical analysis steps that must be done during single-omics analysis before attempting to integrate multi-omics datasets. OmicsNet is used for (**b**) protocol Stage 2, which demonstrates how to use pre-existing knowledgebases to find connections between multi-omics feature lists. OmicsAnalyst is used for (**c**) protocol Stage 3, which shows how to use jDR to perform integrated statistical analysis of multiple omics matrices. Comp, component; TF, transcription factor.

Understanding the main statistical properties and functional themes of each omics dataset makes it easier to pinpoint shared versus complementary trends within the integrated results. In addition, thorough and careful quality checks, data filtering and normalization are prerequisites for robust data-driven multi-omics integration.

In this protocol, we will use a previously published multi-omics dataset collected from human islet and blood tissues sampled from patients undergoing pancreatic surgery[30]. The aim of the original study was to understand the molecular phenotypes of islet tissues along the spectrum of diabetes progression. Bulk RNA sequencing (RNA-seq) and proteomics data were obtained from islet tissues, and lipidomics data were collected from plasma samples. In addition, blood glucose levels including hemoglobin A1c (HbA1c) and oral glucose tolerance test outcomes were available from most patients. We analyze these data in the following three stages:

1. Single-omics analysis with ExpressAnalyst and MetaboAnalyst (Steps 1–31; Fig. 1a)
2. Knowledge-driven multi-omics integration with OmicsNet (Steps 32–51; Fig. 1b)
3. Data-driven multi-omics integration with OmicsAnalyst (Steps 52–71; Fig. 1c)

The following three sections elaborate on some statistical and biological concepts that will be helpful for understanding and executing the protocols.

General workflow for single-omics analysis

To achieve comprehensive molecular characterization, multi-omics studies generally leverage complementary omics technologies. For instance, next-generation sequencing (NGS) is used for genomics, epigenomics or transcriptomics, while mass spectrometry (MS) is employed for proteomics, metabolomics or lipidomics. These omics platforms generate very different raw data (FASTQ files versus MS spectra) and require distinct algorithms for raw data processing. NGS raw data processing tasks have been consolidated into a few well-established pipelines[6,31] while MS spectra processing is still under active development, although substantial progress has been made recently[32–37].

The result from raw omics data processing is a high-dimensional table containing abundance measures for different molecules (transcripts, proteins, metabolites, etc.), which can be analyzed within a consistent bioinformatics workflow involving three major steps: (1) data processing including data filtering, missing value imputation and normalization to improve data quality for downstream statistical analysis and machine learning. Since noise and missing values are characteristics of the underlying analytical platforms, this step generally employs approaches tailored to the target omics types; (2) comparative analysis, also known as differential analysis, aiming to identify molecular features that are significantly associated with the phenotypes of interest; and (3) functional interpretation, which connects changes of individual molecules with biological activities, often through enrichment analysis developed for different omics types[38–41]. Applying the above three steps enables a good understanding of individual omics layers and provides a solid foundation for multi-omics integration.

Using biological networks for knowledge-driven multi-omics integration

Networks are natural representations of our knowledge. When used appropriately, networks can greatly facilitate information uptake and reasoning[2,13]. Biological networks describing gene regulations, protein–protein interactions (PPIs) or metabolic reactions are fundamental for multi-omics integration and systems biology[2,42]. A common practice is to project molecules of interest into a network context and study their connectivity patterns. For instance, single-omics analysis allows identification of significant features consisting of lists of molecules ranging from tens to low hundreds. These signatures can be mapped into a suitable biological network such as PPI network to identify key functional modules or activity hotspots within this well-defined context.

A key concept in constructing informative networks is to strike a balance between 'signals' and 'context' to facilitate data understanding. We describe two common strategies. The first, context creation through signal enrichment, is a classical approach that uses tens to hundreds of significant features or seeds to query a comprehensive knowledgebase to obtain their individual interaction partners from the same or different omics layers. These connections are

subsequently merged to form a large network for visualization and functional analysis. The second strategy, context creation through signal distillation, is suitable when we detect a large number of signals (~1,000s of significant features), which already define a sufficient context by themselves. Applying the classical approach tends to generate large, densely connected networks that are hard to view and hard to understand (known as 'hairball effects')[43]. Instead, the strategy is to refine this context by identifying known relationships among those signals. After creating a suitable network context, a variety of algorithms can be leveraged to reveal functional insights[16,25].

Using DR for data-driven multi-omics integration

Data-driven approaches aim to perform de novo identification of patterns shared across different omics layers[44]. To achieve this, they need to address the issue called 'the curse of dimensionality'. Most computational models require many more samples (n) than features (p) for parameter estimation[18]; however, omics data are characterized by small n and large p ($n \ll p$), and multi-omics data only exacerbate this issue. Very few computational approaches can be directly applied in such scenarios. Among them, dimensionality reduction (DR) approaches show the most promising performance in recent benchmark studies[17,45]. Here, we introduce several basic concepts that are necessary to understand and interpret the results from DR approaches used in this protocol.

The primary objective of DR is to compute a low-dimensional representation that captures the main characteristics of the high-dimensional data. There are two important assumptions: (1) the high-dimensional data contains redundant information so it can be greatly compressed without losing important messages and (2) typical statistical summaries such as variance for single-omics data (X) and covariance for multi-omics data ($X1, X2$ ....) or between omics data and phenotypes ($X1, X2, \ldots Y$) can adequately capture key characteristics of the data. Both assumptions generally hold well—omics data contain lots of redundant information in the form of interdependent, co-expressed or co-abundant features (i.e., molecules involved in the same biological processes) that are often linearly correlated. Therefore, it is reasonable to expect that the total variability in omics data can be explained by a much smaller set of components (note, some DR methods also use the term 'factors'). Despite differences in the mathematical implementation of different DR methods, their results can be interpreted using a common framework, as they all compute loadings (feature coefficients) that are multiplied with the omics data to obtain scores (samples in the low-dimensional space defined by the components). We can annotate scores with different metadata labels to detect whether any of the main trends correspond to sample descriptors such as age, sex, diagnosis, etc. Once we have identified the components with interesting patterns, we can inspect the corresponding loadings to see which features are influential, and further explore their functional implications.

Alternative methods

The most common alternatives for Stage 1 (single-omics workflow) are custom scripts based on R packages such as limma, DESeq2 or MetaboAnalystR[32,46,47]. If researchers are already very experienced with single-omics data processing, differential analysis and functional analysis, then Stage 1 can be skipped, as the results (which are the inputs for Stages 2 and 3) are available for direct download. If researchers choose to skip Stage 1, we still recommend reading through the steps as some processing and analysis methods were specifically designed for downstream multi-omics integration, and some methods and results are referenced in Stages 2 and 3 of the procedure. There are several other tools for knowledge-based multi-omics integration, including PaintOmics, MergeOmics and several Cytoscape plug-ins[15,48]. Compared with OmicsNet, these tools have fewer multi-omics knowledgebases, fewer options for network layouts and no support for three-dimensional (3D) network visualization[16]. Most currently available methods for data-driven multi-omics integration are available as R or Python-based packages. These tools generally require programming skills to be used effectively and are more suitable for data scientists[9]. A few web-based tools are available for data-driven integration, including 3Omics and MiBiOmics[49–51]. They offer relatively simple visual exploration and correlation analysis across different omics layers. More recent multivariate statistics and

machine learning approaches for multi-omics integration are not well supported[52]. The fast-evolving nature of these methods and their open-ended analyses have made it extremely challenging to distill the complex procedures into intuitive, interface-driven tools[8,10]. Overall, the main decision for researchers is whether to use custom R or Python scripts or graphical user interface (GUI)-based software such as OmicsNet and OmicsAnalyst for multi-omics integration. In addition to being more user friendly, GUI-based tools usually offer interactive visualizations to help navigate complex multi-omics results. The main downside of GUI-based tools is that they are typically designed for the most simple and common experimental designs and may not have enough flexibility to support all analysis objectives for a more complex dataset.

Advantages, limitations and additional applications

Multi-omics data analysis is fast evolving, with new methods being developed daily. The key bottleneck is how to make these methods accessible to many bench researchers who are not experts in statistics and programming. The Analyst software suite helps researchers navigate this complex landscape by making expert-designed omics analysis workflows accessible via web-based interfaces, allowing researchers to process their data without any coding. In addition, the tools couple complex statistical results with curated knowledgebases and powerful visual analytics to make the results easier to interrogate and interpret. These tools share common interface design concepts, making it easier for researchers to become familiar with multiple tools. This protocol outlines how to use the Analyst tool suite for multi-omics integration, while introducing established concepts and common practices in the field.

This protocol also has several important limitations. First, it does not cover integrating data from genomics and epigenomics studies, although OmicsNet, miRNet or mGWAS-Explorer all accept single-nucleotide polymorphism (SNP) input. Interested users are referred to several recent publications[53–55]. Second, this protocol does not cover deep learning methods that are complementary to the DR approaches used in this protocol in terms of predictive performance and capability of capturing nonlinear relationships[56]. Additionally, this protocol does not cover single-cell multi-omics and spatial multi-omics, which are poised to play important roles in future multi-omics studies[57,58]. Finally, all of the tools covered in this protocol have limitations in the types of input data that they accept, as it is not possible to support all ID types for all omics platforms.

Other than the topics covered in the protocol, OmicsNet and OmicsAnalyst can be used to analyze paired miRNA and mRNA expression data. In this case, OmicsAnalyst can be used to construct miRNA–mRNA correlation networks directly, or to first identify lists of miRNAs and mRNAs of interest and upload them to OmicsNet to identify known miRNA–target interactions. The Analyst tool suite including ExpressAnalyst, MetaboAnalyst, MicrobiomeAnalyst and OmicsAnalyst can also be applied to analyze high-dimensional data that does not fall under the classic definition of 'omics' data. While they are designed to recognize specific ID types, they can accept any data tables for statistical analysis and visualization, with the caveat that functional analysis will not be possible as the data have not been annotated by the internal libraries. For example, untargeted chemical data is increasingly collected alongside metabolomics or transcriptomics data during exposomics studies and could be integrated using the strategy described above. Also, while OmicsNet was primarily designed to integrate multiple lists of omics features, users can also upload graph files created using other tools for visual exploration using its powerful two-dimensional (2D) and 3D network visualization system.

Experimental design

Coverage and quality

Researchers should be aware of the inherent limitations of different omics technologies during study design and data analysis (Box 1). A critical consideration is the difference in the quality and coverage of molecules that can be measured from different omics layers[8,18,59]. For instance, transcriptomics can provide high-quality unbiased measurements of expression profiles for 10,000s of genes across different tissues and organisms[60–62]. The coverage of proteomics has been substantially expanded in recent years but is still less than 10,000 in most datasets. Improving the coverage and quality of protein quantification remains an active research area.

BOX 1

Experimental design considerations

Different experimental designs put clear constraints on the strategies that are appropriate for multi-omics integration, and impact how we should interpret results[59].

Sample matching

Different omics data should ideally be collected from the same set of biological samples. The requirement is essential for data-driven integration, which aims to identify shared patterns and correlated molecular features directly based on the omics data themselves. Knowledge-driven integration is much more flexible as omics signatures can be obtained from related but independent samples. Much existing knowledge is about interactions at molecular and cellular level, for example, PPIs or metabolic reactions. It is important to keep in mind that different omics layers operate at very different time scales, which could have a large effect on the measured correlations[83]. The relevance of this knowledge to our multi-omics studies depends on how physically and temporally close they are to each other, and the overall analysis context.

Sample size

Certain methods do not give robust results with smaller sample size[21,59]. In general, we recommend at least 20 samples per group for performing DR analysis. In addition, differences in statistical power based on sample size relative to the variations associated with the underlying omics platforms should be considered when comparing and integrating results. One omics layer having more significant features does not necessarily mean that that it is more biologically relevant, but rather could reflect superior technical performance of the platform.

Feature annotation

Different omics types and different species have different levels of functional characterization. In addition, some platforms have systematic bias in their coverage. MS-based omics platforms are more prone to this issue compared with NGS-based platforms[8,84]. For platforms with limited coverage, considering the 'background universe' of detected features can partially address the issue. For instance, OmicsAnalyst automatically uses all features to filter the functional libraries before performing analysis. This is not possible when only signals (significant features) are provided which is the case for OmicsNet. For poorly annotated platforms, functional analysis is difficult, and we may be limited to identifying biomarkers statistically.

In comparison, metabolomics and lipidomics currently provide only limited coverage (tens to hundreds) with targeted approaches[4], while global or untargeted approaches do not allow robust identification and quantification for the 1,000s of features detected by the current platforms[63].

Sample size

Different statistical methods require different sample sizes to achieve robust results (Box 1). In general, we recommend at least five samples per experimental group for single omics analysis (Stage 1) and knowledge-based integration (Stage 2), and a minimum of 20 samples per group for data-driven integration (Stage 3). Each tool explicitly supports different species for functional analysis. MetaboAnalyst currently supports pathway analysis for 26 species. ExpressAnalyst has annotation libraries for 29 species. OmicsNet supports ten model species, as it relies on pre-existing knowledgebases for integration. If users have multi-omics data from non-model species, they can still perform statistical analysis for single-omics data as well as data-driven multi-omics integration using OmicsAnalyst.

Materials

Equipment

Computer requirements

- Browser requirements: all tools have been tested on major modern web browsers. For the best results, we recommend Google Chrome 92+, Firefox 92+, Safari 12+ and Microsoft Edge v93+. JavaScript must be enabled in your browser
- Internet connection requirements: a fast connection (>1 MB/s) is highly recommended
- Hardware requirements: a computer with >4 GB of memory and screen resolution of at least 1,200 × 800 is preferred. For best experience, use a computer screen that is at least 15 inches

Data files

A typical multi-omics study will generate a large amount of data at different stages. Among them, two types of tables, metadata and omics data, are essential for conducting multi-omics analysis. A metadata table contains various experimental factors or phenotype descriptors for individual samples. Large studies often come with complex metadata that includes both discrete (i.e., diagnosis, technical batch, sex) and continuous (i.e., age, body mass index (BMI)) variables. An omics data table contains abundance information for many (100s–10,000s) molecular features obtained from a particular omics layer, representing a high-dimensional descriptor of the same set of samples.

All datasets used in this protocol are from a previously published study that collected multi-omics data from islet tissues (transcriptomics and proteomics) and plasma samples (lipidomics) from donors undergoing pancreatic surgery[30]. Donors were categorized along the continuum of diabetes progression, including those with no diabetes (ND), impaired glucose tolerance (IGT), type 3c diabetes (T3D, impaired glucose response caused by pancreatic injury/inflammation), and type 2 diabetes (T2D). In total, there were 135 donors; however, not all omics types were collected from all donors. The range of experimental designs and sample sizes makes this a good case for evaluating appropriate analysis workflows based on specific dataset parameters. Data from this study include the metadata table and three omics data tables. More details on how these types of files should be formatted for the web tools are given below.

- Metadata table. A single metadata table is used to annotate all other uploaded omics data. The first column should be sample IDs. Each sample ID must be unique and contains a combination of letters, numbers and underscores (no spaces or special characters). Additional columns should contain metadata variables (discrete or continuous). To maximize data usage, missing values are tolerated on data upload. Instead of removing any sample with a missing value for any metadata variable, a complete case analysis will be performed before conducting the statistical analysis. This means that, in downstream steps, after the user decides which factors to consider in their analysis, the data will be filtered to exclude those samples with missing value for only the selected metadata variables, and only for that specific analysis
- Omics data table. The omics data tables contain the abundance values of molecular features measured for the same sample IDs. Since omics data is high dimensional, they are usually saved in the 'long' format with samples in columns and features in rows. Note for targeted metabolomics data with ~100 features, it is also common to use the 'wide' format (i.e., samples in rows). Missing values are generally allowed in omics data. Depending on omics types, different data filtering and missing value estimation algorithms are offered in the data processing stage to help clean the data and to improve data quality

Equipment setup

(Optional) Download the example datasets

Users can directly use the 'Try Examples' button to select and submit the data for analysis. Alternatively, users can download the datasets and re-upload them to the appropriate module of each tool. These datasets are available here:

- Metadata: https://www.xialab.ca/api/download/NP_multiomics/metadata.csv
- Transcriptomics: https://www.xialab.ca/api/download/NP_multiomics/raw_rnaseq.txt
- Proteomics: https://www.xialab.ca/api/download/NP_multiomics/raw_proteomics.txt
- Lipidomics: https://www.xialab.ca/api/download/NP_multiomics/raw_lipids.txt

Procedure

Stage 1: single-omics analysis with ExpressAnalyst and MetaboAnalyst

● **TIMING ~45 min**

1. Starting up. To keep everything organized, create a folder 'Multiomics_protocol' for all results generated. In this stage, the transcriptomics data (raw_rnaseq.txt) will be analyzed

first, then the proteomics (raw_proteomics.txt) and finally the lipidomics (raw_lipids.txt). Some of the results from Stage 1 will be used as inputs for integrative analysis during Stages 2 and 3. The protocol will indicate whenever results should be downloaded and saved.

Transcriptomics data analysis

2. Go to the ExpressAnalyst homepage (www.expressanalyst.ca) and click 'Start Here' to enter the module view page. There are five different modules for different types of data inputs, including FASTQ files, a gene list, a single gene expression table and multiple gene expression tables, as well as database queries for non-model species RNA-seq analysis using our Seq2Fun pipeline[64,65]. Since there is a single RNA-seq table, click the middle 'Start Here' button below the single table icon within the 'Statistical and Functional Analysis' box. There are two options for data formatting and uploading. Users can either download the protocol datasets and perform all upload steps manually or access each dataset using the 'Try Examples' button. We suggest that users manually download and upload at least one dataset to get familiar with this important step.
3. Data formatting. Upload the metadata file and the omics data as two separate files. Please refer to the 'Data Files' subsection of the 'Materials' section above for details on how the metadata and omics data table should be formatted. Open the 'raw_rnaseq.txt' and 'metadata.csv' using a spreadsheet editor such as Excel to familiarize yourself with the correct formatting.
4. Data upload. First, enter the following data information to inform ExpressAnalyst to use the appropriate workflows:
   - Set organism to 'H. sapiens (humans)'
   - Leave analysis type as 'Differential Expression'
   - Set data type to 'Counts (bulk RNA-seq)'
   - Set ID type to 'Ensembl Gene ID'
   - Choose the RNA-seq count table for the 'Data File' upload, leaving 'Metadata included' unchecked
   - Choose the metadata table for the 'Metadata File'
   - Click 'Submit' to start uploading. A warning should appear saying that two samples in the metadata file were not found in the omics data file and were removed. This is because the metadata file includes all samples across the transcriptomics, proteomics and lipidomics data, but not all samples are present in each omics data. Samples that do not have both metadata and omics data are removed upon data upload. Once the upload has been finished, click the 'Proceed' button at the bottom of the page
     ▲ **CRITICAL STEP** If the feature IDs are not recognized by the tools, the data can still be uploaded and processed (by keeping the option as 'not specified'). However, downstream functional analyses will be disabled since IDs are necessary to map between uploaded features and built-in libraries for functional annotation.
     ◆ **TROUBLESHOOTING**
5. Data quality check. This page allows users to examine the uploaded omics data and metadata and to perform manual correction if necessary. The default 'Omics data overview' tab provides both text and graphical summaries of the uploaded RNA-seq data. The summary shows that the number of features was greatly reduced from ~58,000 to ~20,000. This is because the uploaded Ensembl gene IDs were mapped to Entrez gene IDs. Many Ensembl gene IDs describe novel transcripts that are not well-characterized protein-coding genes and that do not have functional annotations, and therefore they do not have a corresponding Entrez ID. From the box plot, it is clear that the data are raw counts because they include many counts values in the 100s and 1,000s. Click on the 'Metadata overview' tab and make sure that each metadata has been classified correctly: 'Diagnosis' should be 'Discrete' and the other four should be 'Continuous'. It is also important to make sure that the 'Diagnosis' classes are ordered according to diabetes progression in all downstream plots. Click the 'Edit metadata column', navigate to the 'Order (factor-level)' tab, sort the classes to 'ND-IGT-T3D-T2D', and click 'Update'. Now the data is ready for normalization. Click 'Proceed'. A warning will pop up saying that there are some missing values in the

metadata, and if a variable with missing values is selected in the downstream statistical analysis, samples with missing values for this variable will be filtered out (i.e., complete case analysis). Click 'OK'.

6. Filtering and normalization. For high-dimensional omics data analysis, it has been well established that removing features with very low abundance or low variance across all experimental conditions can substantially improve statistical power while retaining most of the informative results[66]. Leave the filtering as the defaults (4 for 'Low abundance' and 15 for 'Variance filter'). There are four methods provided for normalizing RNA-seq count tables. 'Log2-counts per million' normalizes based on sequencing depth and then transforms to the log scale while the other three methods additionally correct for biases introduced by highly abundant transcripts[67]. Select the 'Relative log expression (RLE) normalization' method for normalization and click 'Submit'. In addition to correcting for differences in sequencing depth and log-transformation, RLE normalization calculates sample-specific scaling factors based on the geometric mean of each gene across all samples. After normalization, all expression values are more comparable and less than 15—a clear sign of log transformation. Click 'Proceed'.
7. Differential expression analysis. Two tabs are available for differential expression analysis. The default 'Simple Metadata' panel offers common options for specific statistical contrasts and supports up to two categorical factors for convenience and consistency with published protocols. The 'Complex Metadata' panel allows more flexible experimental designs such as analyzing continuous metadata variables such asHbA1c or adjusting for multiple covariates based on the limma R package[47]. Navigate to the 'Complex Metadata' tab since our metadata contains five factors.
8. The first step is to identify genes that are significantly associated with HbA1c, which reflects average blood glucose level over the past 2–3 months. Select 'HbA1c' as the primary metadata and include 'Age' and 'BMI' as covariates. If metadata on 'Sex' were publicly available for these donors, this variable would be included as well. Leave other options as default. Click 'Submit' and 'Proceed'. There should be 928 genes associated with HbA1c. Clicking the image icon in the 'View Details' column for the top ranked genes to view graphical summaries of their distributions in relation to HbA1c. Change the metadata and update to see how the data looks like for other variables.
   ◆ **TROUBLESHOOTING**
9. The next step is to identify significant genes that are differentially expressed between specific diagnosis groups. Navigate back to the 'Differential Analysis' page and click the 'Complex Metadata' tab. Change the primary metadata to 'Diagnosis'. Select 'ND' for the reference group, 'IGT' for the contrast, leave 'BMI' and 'Age' as covariates, click 'Submit' and click 'Proceed'. No significant genes were identified on the 'Select Significant Genes' page for this comparison. Click the 'Previous' button on the left side of the footer and change the contrast to 'T3D' within the 'Complex Metadata' tab. Click 'Submit' and then 'Proceed'. Now there should be 535 differentially expressed genes (DEGs). Navigate back once more and re-do this step for the 'T2D' contrast. This contrast has by far the greatest effect size, with 1,678 DEGs, mirroring what was found in the original study[30]. Click the 'Download' button and save the T2D versus ND comparison results as 'sig_t2d_rnaseq.csv'. Click 'Proceed'.
10. Functional analysis. The 'Analysis Overview' page provides a suite of visual analytics methods that couple interactive visualization with common enrichment analyses to facilitate pattern discovery and data understanding. Overrepresentation analysis (ORA) compares the list of DEGs with the list of genes in each pathway and determines whether there are more genes from that pathway in the list of DEGs than we would expect due to random chance using the hypergeometric distribution. In other words, it computes whether a given pathway is 'overrepresented' in the list of DEGs against a 'background universe', which can be defined as all genes in the uploaded data (default for expression table input) or the gene set library (used for gene list input). Gene set enrichment analysis (GSEA) analyzes the entire list of measured features, ranked by some summary statistics (such as fold changes (FCs) or *t*-statistics), and evaluates whether members of a pathway are concentrated at the extremes of the list (top or bottom). GSEA is computed based on

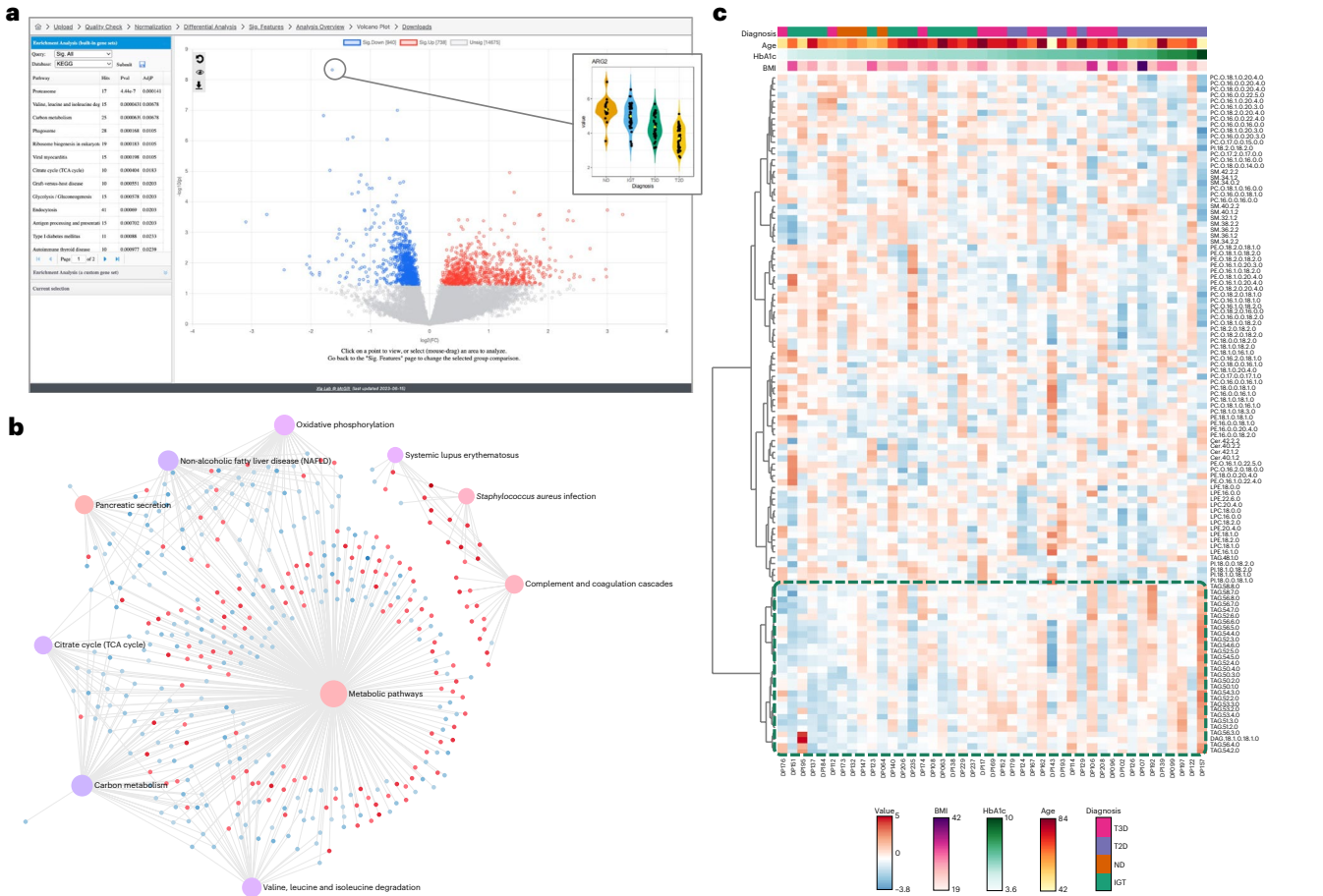


Fig. 2 | Results from the single-omics analysis. a–c, Differential expression and ORA results for the transcriptomics data (**a**), GSEA results for the proteomics data (**b**) and a heat map of lipidomics data, with samples sorted according to HbA1c values (**c**). Features with positive association with HbA1c in (**c**) are outlined in the dashed green box, and heat map cells are colored from red to blue according to lipid abundances.

the fgsea R package[68]. Since the last statistical comparison was 'T2D vs. ND', these are the results that will be analyzed with the visual analytics tools. Leave the 'Background universe' dropdown as 'Uploaded data' and click the 'Volcano Plot'.

11. The volcano plot provides an overview of all genes, with the *x* axis indicating their log2 FCs and *y* axis shows their −log10 adjusted *P* values by default (Fig. 2a). Hover over any data point to show its name and click on it to view its violin plot across different groups. The panel on the left shows the enrichment analysis results based on the DEGs against Kyoto Encyclopedia of Genes and Genome (KEGG) pathways identified in the previous step. Click on the 'Citrate cycle (TCA cycle)' row in the 'Enrichment Analysis' panel to highlight all DEGs belonging to this pathway ('hits') on the volcano plot. Those genes were also displayed on the 'Current selection' panel on the bottom left. Notice that all hits had lower expression in donors with T2D (negative log2 FC) for this pathway (Fig. 2a).

◆ **TROUBLESHOOTING**

12. Sometimes pathway-level results are easier to interpret if the analysis is performed separately on the up- and downregulated transcripts. To find enriched pathways in the list of DEGs that are upregulated in patients with T2D compared with patients without diabetes, go to the top left part of the screen and change the 'Query' from 'Sig. All' to 'Sig. Up' and click 'Submit'. The results table shows that the most significant pathways are related to autoimmune processes, where all autoimmune pathway results are based on the same set of upregulated transcripts. Next, change the 'Query' to 'Sig. Down' and click 'Submit'.

This list of pathways has more mixed biological themes, but many pathways are related to metabolism (proteasome, carbon metabolism, fatty acid metabolism, etc.) and energy generation (citrate cycle, glycolysis, etc.). Spend some time exploring the other visual analytics tools on your own.

13. Download. To save a copy of the normalized data for our downstream analysis, click the 'Downloads' link at the end of the navigation bar at the top of the page. All files, scripts and figures generated during the analysis are available for download. Under 'Files & Scripts', find 'data_normalized.csv' and save it as 'normalized_rnaseq.csv' to distinguish these data from the normalized proteomics and lipidomics matrices that will be created later. Note that there will be many fewer features in the normalized data (~16,000 instead of ~58,000). In addition to the reduction in features after mapping from Ensembl Gene IDs to Entrez Gene IDs (explained in Step 5), the counts table was filtered based on both feature variance and abundance (Step 6).

Proteomics data analysis

14. The next set of steps are to analyze the proteomics data with ExpressAnalyst. Go back to the homepage (www.expressanalyst.ca) and click refresh, this will ensure that ExpressAnalyst will start a new session for the second analysis. Navigate to the module view page and click the 'Start Here' under the single expression table. The workflow through differential analysis is very similar to the RNA-seq analysis since we are using the same module. Repeat Steps 4–9, other than the following differences:
    - Data upload (Step 4). The data type is 'Intensities (proteomics)', the ID type is 'Official Gene Symbol'. Choose the proteomics table for the data file and the same metadata table for the metadata file. The original study used label-free quantification; however, ExpressAnalyst can also be used to analyze data using various quantification methods, as long as the raw data processing has already been completed and the data are summarized as a table of intensities/relative abundances
      ▲ **CRITICAL** Note that, as with the transcriptomics data, ExpressAnalyst will automatically filter the metadata file and proteomics data file to retain only samples that are present in both files. The result of this filtering is reported in one of the notifications after data upload.
    - Quality check (Step 5). There are only two groups of donors (T2D and ND), so it is not necessary to re-order the 'Diagnosis' metadata variable classes. This data does not contain missing values; however, ExpressAnalyst does support missing value imputation for proteomics data using the 'Missing value estimation' tab on this page
    - Filtering and normalization (Step 6). Since there are many fewer proteomics features compared with the RNA-seq data, do not perform any filtering (set both abundance and variance filters to zero). Use the 'Variance Stabilizing Normalization (VSN)' normalization method, which has been shown to perform well for proteomics data[69]
    - Differential expression analysis (Steps 7–9). There are two donors with proteomics but no RNA-seq data, and metadata other than 'Diagnosis' were not provided. Keep 'Diagnosis' as the primary metadata, make sure ND is the reference group and compare T2D with ND with no covariate adjustment since adjusting for age and BMI will reduce our sample size from 10 to 8. Since there are no IGT or T3D donors, we cannot perform those comparisons. You should get 118 significant results for T2D versus ND (no covariate adjustment). Make sure to save the differential expression analysis results table for T2D versus ND as 'sig_t2d_proteomics.csv'
15. Functional analysis. ORA analysis with the volcano plot tool results in lists of pathways with similar biological themes compared with the RNA-seq data; however, none of the adjusted *P* values is significant. GSEA, which uses the entire list of genes/proteins rather than only the DEGs, can identify potential pathways with subtle yet consistent changes when the perturbation of individual genes/proteins are too small to be identified as DEGs[69]. In the 'Analysis Overview', click the 'Enrichment Network' icon.
16. This tool uses networks to organize significant functional analysis results. Pathways are represented as nodes with sizes proportional to the number of features in the pathways and

color gradients based on their *P* values (yellow = least significant, red = most significant). Two pathways are connected if their intersecting list contains at least 35% of the features from the smaller pathway, which can help illustrate functional redundancies and common biological themes across the set of significant pathways. Like the volcano plot, the results from the enrichment analysis (default ORA) are shown on the left pane. Click the 'Type II diabetes mellitus' pathway to highlight its corresponding nodes in the network. Double click this node in the network viewer to view the pathway's significant hits (red = positive log2FC, blue = negative log2FC). Here, this displays the 'SLC2A2' (lower expression in donors with T2D) and 'HK1' (higher expression in donors with T2D) nodes. The network shows that both genes are also involved in the 'Central carbon metabolism in cancer' and 'Carbohydrate digestion and absorption'. 'SLC2A2' is also in the 'Maturity onset diabetes of the young' pathway, and 'HK1' is in several other pathways, including the 'Insulin signaling pathway'.

17. The top ORA results were consistent with previous findings; however, none of the pathways was significantly enriched (adjusted *P* values all >0.05). Since GSEA considers the entire ranked list of features and not just the DEGs, it can sometimes detect significant enrichment when there are coordinated changes in pathway expression levels even when the individual pathway genes were not differentially expressed[39]. In the 'Enrichment analysis' panel, change the 'Type' to GSEA, leave the database as 'KEGG', select 'Fold change' for 'Rank (GSEA)' and click 'Submit'. When we did this, there were ten pathways with adjusted *P* values less than 0.05. Note that GSEA uses permutations to compute *P* values and will therefore produce slightly different results each time.
18. The network can be further customized to make it more informative. In the panel above the network, change the 'View' to 'Bipartite network' to expose the individual genes/proteins within each pathway. Change the 'Background' to 'White'. It is now apparent that the immune-related pathways (complement and coagulation cascade; *Staphylococcus aureus* infection) contain mostly upregulated proteins while others (citrate cycle (TCA cycle), oxidative phosphorylation, etc.) contain more of a mix (Fig. 2b). This is consistent with the functional analysis of the RNA-seq data, which increases confidence that both omics data are capturing robust biological trends.

Lipidomics data analysis

19. The final steps in Stage 1 will use MetaboAnalyst to process and analyze the lipidomics data. Go to the MetaboAnalyst homepage (www.metaboanalyst.ca) and then click 'Click here to start' to navigate to the module overview page. Different analysis modules are organized based on their input data types. Select 'Statistical Analysis [metadata table]' from the third row.
20. Data upload. Keep the data type as the default 'Concentrations' and study design as 'Multiple factors / covariates'. Change the data format to 'Samples in rows'. Select the lipidomics data file and the metadata file and click 'Submit' to upload. Alternatively, choose the last example data in the 'Try our test data' panel and click 'Submit' below. This will navigate to the 'Data Integrity Check' page, which contains a summary of the uploaded dataset. If there are missing values in the omics data, you can click the 'Missing Values' button to access different methods for imputation. Since no missing values are detected in this case, click 'Proceed'.
21. Metadata check. Ensure that each metadata variable has been classified correctly, with diagnosis as categorical, and age, HbA1c, oral glucose tolerance test and BMI as continuous. MetaboAnalyst will automatically order categorical classes alphanumerically, which means that for diagnosis, the order will be IGT, ND, T2D and T3D. It makes more sense if the class order corresponds to increasing diagnosis severity. To manually adjust this, click the 'Edit' link in the 'Diagnosis' row. Navigate to the 'Order (factor-level)' tab, re-order to ND, IGT, T3D, T2D and click 'Update'. Click 'Proceed', a dialog will pop up giving some information on how MetaboAnalyst handles nonstandard metadata variables. Click 'Confirm'.
22. Filtering. As with transcriptomics and proteomics data, metabolomics and lipidomics data can benefit from filtering before statistical analysis if there are many features. Here, there are only 116 features, which is a relatively small number. Drag the 'Percentage to filter out' slider on the right to 0% and click 'Submit' and then 'Proceed' to avoid filtering out any features.

23. Normalization. Sometimes it is not clear whether an omics feature table is normalized. Leave the sample normalization, data transformation and data scaling as 'None' and click 'Normalize'. Then, click 'View Result'. Since no normalizations were performed, the 'Before Normalization' and 'After Normalization' plots should look the same. Both the feature view and sample view show that the data covers a wide range of values, making them difficult to visualize and to compare, and that many features have right-skew distributions. Close the normalization view dialog.
24. First, transform the data to better visualize the distributions by selecting 'Log transformation' in the 'Data transformation' section. Click 'Normalize' and click 'View Result'. It is now much easier to visualize the distribution of individual features on a log scale, which are all approximately normally distributed. Navigate to the 'Sample View' tab. After log transformation, the sample-level distributions are more similar to each other, but not exactly lined up and could benefit from sample normalization. Close the dialog.
25. Keep 'Log transformation' selected, select 'Normalization by median' in the 'Sample Normalization' section, click 'Normalize' and click 'View Result'. Navigate to the 'Sample View' tab. The sample-level distributions are now aligned. In general, we recommend applying simpler methods first (i.e., 'Normalization by median' or 'Normalization by sum' and only proceeding to more complex methods if needed (i.e., supplying additional information such as sample weight or references). Close the normalization dialog and click 'Proceed'.
26. Differential analysis. There are many analysis options after the normalization page. This protocol only uses the 'Linear Models with Covariate Adjustment' tool to detect a list of significant lipids. Please see our other protocols for more information on the other analysis options, and other modules in MetaboAnalyst[22,23]. Click the 'Linear Models with Covariate Adjustment' link.
27. This page should look familiar to the differential expression analysis page in ExpressAnalyst as they share the same statistical method. First, set the primary metadata to 'Diagnosis', select 'ND' as the reference group, and select 'IGT' for the contrast to find metabolites with significant differences between 'ND' and 'IGT'. Next, select age and BMI as covariates and click 'Submit'. One significant feature is detected with the default false discovery rate (FDR) cut-off of 0.05, suggesting subtle differences. For demonstration purposes, change 'P-value cutoff' from 'FDR' to 'Raw' and click 'Submit'. There are four significant lipid features, three of which were only significant because age and BMI were accounted for. Click individual features within the plot to see sample-level data points and update the metadata within the dialog to visualize their associations with the different metadata variables.
28. Next, change the contrast group to 'T3D' and click 'Submit', and then to 'T2D' and click 'Submit'. There are seven and four significant lipids respectively. The results also show that there are eight lipids that would have been significant if no covariates were considered. Normally, the next step would be functional analysis; however, these features are annotated with very general classes that are difficult to map to specific pathways, as is often the case with lipidomics data. For these reasons, functional analysis will yield no results, or only those with very low confidence. The other data in this study are from the pancreas, which is an extremely inaccessible organ, while the lipids were measured in plasma. Since plasma is easily accessible, certain lipid features could theoretically be used as biomarkers for islet dysfunction. These ideas will be explored further in Stage 3 of this protocol.
29. The last statistical analysis for this data is to look for lipids associated with HbA1c. Navigate back in your browser. Change the primary metadata to 'HbA1c' and click 'Submit'. There should be a total of 45 significant features.
30. Heat map view. Since there are a small number of lipids, the data for all features can be visualized in a single heat map. Go to the navigation panel on the left and click 'Multi-factors'. This will navigate back to the visual analytics overview. Click the 'Hierarchical Clustering and Heatmap Visualization' link. Leave most parameters as the defaults, except for change the 'Sample Arrangement' to 'HbA1c', and select the 'Show names' check box for 'Row option'. This will generate a heat map of all the data, with lipids hierarchically clustered and samples arranged from lowest to highest HbA1c values.

Some lipids appear related to HbA1c, with the group of primarily 'TAG' (triacylglycerol) lipids at the bottom having generally higher abundances in samples with higher HbA1c (Fig. 2c). However, the patterns are more subtle than they were for the proteomics and transcriptomics data.

31. Download the normalized data. Click the 'Download' link, near the bottom of the left-side panel. Find the 'data_normalized.csv' table and save it with a more informative name such as 'normalized_lipids.csv'.

Stage 2: knowledge-driven integration with OmicsNet

● **TIMING ~30 min**

▲ **CRITICAL** In Stage 2, the significant features identified from the statistical analysis of the transcriptomics and proteomics data[30] in Stage 1 will be uploaded to OmicsNet for integration. Only the transcriptomics and proteomics features are integrated with OmicsNet for two reasons. First, most lipid features are not well characterized in the literature, so there is a lack of prior knowledge to link specific lipids to other molecular features. Second, the transcriptomics and proteomics data were measured in the same islet tissues while the lipidomics data were measured in plasma. Network-based analysis is highly dependent on the data and analysis context (as discussed further in the 'Introduction').

32. Starting up. Go to the OmicsNet homepage (www.omicsnet.ca) and click 'Click here to start' to enter the data upload page. OmicsNet accepts multiple molecule types (metabolites, miRNAs, transcription factors, proteins, mRNAs, liquid chromatography–mass spectrometry peaks, microbial taxa and SNPs), all of which are lists of features except for the 'Graph File', which allow users to quickly re-visualize the previously created networks saved as common network files.
33. Uploading lists of molecules. Open the 'sig_t2d_proteomics.csv' file, which contains 118 Entrez IDs with adjusted *P* values <0.05. Click the 'Proteins' input type in OmicsNet. Keep the organism as human and set the ID type to 'Entrez ID'. Copy and paste the list of significant IDs into the text input and click 'Upload'. Next, select the 'Genes/mRNAs' input type and 'Entrez ID'. Copy and paste the 1,678 significant mRNA Entrez IDs from the 'sig_t2d_rnaseq.csv' file. OmicsNet accepts up to five different types of lists for network creation; however, only two lists will be integrated in this case. Click 'Proceed'.
34. Database selection. In this page, the two input lists (called 'seed features/nodes') are summarized on the left panel. The center shows all knowledge bases available for network creation. OmicsNet currently has 14 databases that contain known or predicted relationships between common molecules. As databases are queried with lists of features, the resulting networks will be summarized in the lower 'Individual Omics Networks' section. Start by making sure that the 'mRNA' list is highlighted on the 'Input List(s)' panel on the left-hand side. Click the 'Protein-protein' tab in the middle 'Database Selection' panel and select the STRING database to find known interactions between the uploaded molecules using the popular STRING database[70].
35. Building subnetworks from individual omics list. The mRNA input list alone contains >1,600 features, which should be able to provide sufficient context to understand functional themes. Check the box next to 'Add edges only' (i.e., 'context creation through signal distillation' strategy) and click 'Submit' to query the database. The result will be shown on the 'Individual Omics Networks' section in the lower half of the center pane. The network should have a size of '789 – 3807 – 742' corresponding to the numbers of nodes, edges and query seeds, respectively. Next, select the 'Proteins' input list from the 'Input list(s)' panel on the left-hand side, select the STRING database, make sure 'Add edges only' is selected, and click 'Submit'. This network should have a size of '230 – 513 – 56'. Now, both the mRNA and protein subnetworks are summarized in the network table.

    ▲ **CRITICAL STEP** If multiple uploaded lists share the same ID system (for example, mRNAs/genes, proteins and transcription factors can all be annotated with entrez IDs) and the 'Add edges only' option is selected, OmicsNet will first merge those input lists and automatically search for connections using the merged pool to help rescue those orphan nodes that have no direct interaction partners in their own list but do have connections to molecules in

other omics lists. The resulting networks are usually larger than the networks created from individual list separately.

◆ TROUBLESHOOTING

36. Multi-omics network construction. Click the 'Proceed' button at the bottom of the page to merge the individual networks based on their shared nodes. The resulting networks are shown on the next page. The first row (subnetwork1) should have 667 seed nodes (mRNA/protein) and 1,822 edges. 'All nodes' should be the same as the number of seed nodes since only connections between the uploaded lists were added, and not additional interacting proteins.
37. Overview of multi-omics networks. The center of the page is a table displaying the list of multi-omics networks sorted based on their sizes—typically consisting of a large subnetwork ('subnetwork1') and several smaller subnetworks. The left-hand panel contains various 'Network Tools' that can be used to further refine the networks to reduce network size while retaining important nodes or structures for visualization. Please refer to our previous protocol, OmicsNet publications, and tutorials (www.omicsnet.ca/OmicsNet/docs/Tutorial.xhtml) for more details[13,16,25,29]. Click 'Proceed' to enter the network viewer page. A dialog will pop up asking users to choose between 2D or 3D network visualization. Leave '2D visualization' selected and click 'Proceed'.
38. Network visualization and customization. The multi-omics network ('subnetwork1' by default) is shown at the center of the page, with gene nodes colored in gray and protein nodes in salmon. If a node is shared by multiple input lists, a pie chart will be used to indicate the multi-origin of the node. In this case, these nodes are shown as half gray and half salmon. The size of each node corresponds to its degree, which is the number of direct connections it has to other nodes in the network. By default, all the seed nodes are highlighted with a blue halo. Since only edges were added in our case, every node is a seed node so highlighting can be turned off without losing any information. Find the icon that looks like a sun (with a tool tip 'Highlight seed nodes') in the vertical toolbar and click it. Go to the top horizontal toolbar and change the 'Background' to 'White'. The mRNA nodes have a very similar color to the edges. Click the colored box beside 'mRNA' in the top left corner and change it to a slightly darker gray to help distinguish between the nodes and the edges.
39. Network layout. The network contains many overlapping nodes and edges, making it difficult to read. Use the top horizontal toolbar to change the 'Layout' to 'Graphopt Layout' to move some groups of densely connected nodes closer to the outside of the network. Next, change the 'Layout' to 'Force Atlas'. This takes the existing structure and applies an algorithm to further spread-out nodes that are close together. Note the individual nodes and edges are now much easier to see. Select 'Label' from the 'Styling' dropdown, go to the 'Display' tab, and change the display option to 'Hide'.

◆ TROUBLESHOOTING

40. Browse the 'Node Explorer' table to look at the most central nodes (i.e., those with high degree or betweenness values). There may be some key features of interest based on the specific research context that we may want to highlight. Here, diabetes researchers identified several nodes of interest in the network based on their expertise. BAG3 regulates cortical actin dynamics[71], an important part of insulin secretion[72]. ATP5F1A is a critical subunit of mitochondrial ATP syntheses, another key aspect of glucose-stimulated insulin secretion that has previously been observed to have altered expression in diabetic mice[73]. Finally, BRCA2 is most known for its links to breast cancer; however, it also has a role in metabolism, including an association with circulating insulin levels[74]. To highlight these features of interest, click the 'Batch Selection' header in the right-hand panel. Enter 'BAG3', 'ATP5F1A' and 'BRCA2' as three new rows, and click 'Highlight'. Use the network to see which other proteins interact with the highlighted nodes. Recall that all nodes in our network were differentially expressed/abundant between donors with and without T2D in either the transcriptomics or proteomics data. This type of targeted analysis, where we focus on features with well understood and characterized functions and relationships, can help us interpret the significance of other, physically interacting features that may be less well known. When you are finished, click the 'Reset' button in the 'Batch Selection' area to remove the highlights.

41. Manual refinement. The automatic layout algorithms are typically 90% of the way to an informative network, and most networks can benefit from some small manual adjustments to fix overlapping labels or nodes. In this network, there are a few spots where nodes are overlapping, making them difficult to distinguish. Find a node that could benefit from a little more space. Hover your mouse over the node, click and drag slightly. Continue until you are satisfied with the appearance of the network. Then, hover your mouse over the vertical toolbar until you find 'Apply edge bundling', and click this icon. A pop-up will warn you that manual node adjustments cannot be performed after edge bundling. Click 'Confirm'.
    ▲ **CRITICAL STEP** We suggest using automatic network layouts while initially analyzing the data, and reserving detailed, labor-intensive adjustment for networks that you intend to share, once you are finished exploring the data. These manual adjustments are much easier to make with a mouse compared with a laptop trackpad.
42. Module identification. Go to the 'Module Explorer' panel on the right side of the page, make sure the 'InfoMap' algorithm is selected and click 'Submit'. This will analyze the network by searching for densely connected subnetworks or modules. The InfoMap algorithm detects community structure by compressing the description of information flows based on random walks[75]. There are 28 modules detected, sorted by size. The next step will focus on the seven largest modules and try to understand their functional significance.
43. Module annotation. Select the checkbox next to 'Module 0' to highlight its nodes within the network. Next, go to the Function Explorer panel (directly above the Module Explorer), change the query to 'Highlighted nodes', change the database to 'GO:BP' (Gene Ontology: Biological Process) and click 'Submit'. This group of nodes is significantly enriched with the 'Ribosome biogenesis' GO term. Change the 'Database' to 'GO:CC' (Gene Ontology: Cellular Component) and click 'Submit'. The top adjusted *P* value is 1.01×10^{-48} for the 'nucleolus'—a very strong signal. It makes sense that distinct cellular components are highlighted in our network, given that STRING is based on PPIs. Unselect 'Module 0' inside of the 'Module Explorer'.
44. Repeat Step 43 for Modules 1–5. The main GO terms are summarized in Table 1, lightly edited for length and clarity. Note that in this case, ORA was performed on a combined gene and protein list, which represents one of several strategies for multi-omics functional analysis (Box 2).
    ▲ **CRITICAL STEP** Make sure that you understand which nodes are being used for enrichment analysis, and the potential biases used to define them. It is valid to use the Function Explorer to annotate our network with biological themes. However, keep in mind how the network was created, especially in cases when new nodes have been introduced to the network. Adding nodes based on prior knowledge (i.e., molecular interaction databases) greatly increases the chances of finding significant enrichment results. This should inform your interpretation of the results.
45. Module highlighting. Finally, adjust the color annotations to highlight the top seven modules simultaneously. First, make sure that no modules are selected in the 'Module Explorer'. Go to

Table 1 | Six largest modules from Steps 42–47 and their top functional annotations

ID	Size	*P* value	GO:BP	GO:CC	Color Hex
0	58	9.74×10^{-21}	Ribosome biogenesis	Nucleolus	#66B33E
1	29	5.9×10^{-9}	DNA damage response	Proteasome complex	#006F49
2	22	3.77×10^{-6}	DNA metabolic process (DNA damage response)	Nucleoplasm	#01A9B5
3	21	4.44×10^{-9}	Translation	Ribosome	#3C46BD
4	20	3.56×10^{-6}	Aerobic/cellular respiration	Mitochondrial matrix	#8A2FE4
5	18	9.8×10^{-4}	Protein folding	Nuclear pore	#C35CA5

ID = module number, size = number of features in the module, *P* value = statistical significance of each module, produced by the 'InfoMap' algorithm, GO:BP = Gene Ontology Biological Process gene set name, GO:CC = Gene Ontology Cellular Component gene set name, Color Hex = color of each module specified in Step 45.

BOX 2

Functional analysis for multi-omics data

A typical single-omics workflow performs functional analysis after statistical analysis by comparing lists of significant features or all ranked features to functional libraries. For multi-omics data, we may take three general different approaches.

Independent statistical and functional analysis

Statistical and functional analysis is performed separately for each omics type and results are qualitatively compared at the end. This is the most flexible and straightforward approach and is demonstrated in Steps 10–12 and 15–18 in Stage 1 of the procedure, and in Step 61 in Stage 3.

Independent statistical analysis and integrated functional analysis

Some enrichment libraries support multi-omics queries. For example, KEGG and Reactome pathways include both genes/proteins and metabolites. In these cases, lists of significant features can be directly merged for functional analysis using ORA. This can improve statistical power, as we can use the joint list to maximally detect impacted biological processes. However, it is also the least flexible approach as it is only possible for specific combinations of omics types. This approach is demonstrated in the functional analysis of the combined list of mRNAs and proteins for each module in Steps 43–44 in Stage 2 of the procedure.

Integrated statistical analysis with independent functional analysis

Statistical analysis is performed across omics types, and features are prioritized based on the integrated results. These prioritized features are subject to functional analysis using single-omics functional libraries. For example, both MCIA and DIABLO compute a set of new components that maximize covariance[20,45]. We can then perform functional analysis on each set of loadings independently to understand main biological functions underlying the covariance. This approach is demonstrated in Steps 62–63 in Stage 3 of the procedure.

the top left corner, click the salmon-colored box next to 'Protein' and change it to black. Next, click the royal blue colored box at the top of the vertical toolbar, paste '#66B33E' into the text input, click 'Choose' and select Module 0 from the 'Module Explorer'. You should see that the Module 0 nodes are now highlighted with a green border. Now, leaving Module 0 selected, click the color picker again, paste '#006F49', click 'Choose' and select Module 1. Continue for Modules 2–6. We suggest using the color hex codes in Table 1, which were chosen using a color-blind accessible palette generator.

46. Network download. Now that each module has been annotated with functional themes, we can save an image of this network and combine all the information. Go to the 'Download' dropdown in the horizontal toolbar, select 'PNG Image', leave the scaling factor as 1×, and click 'Save'. This figure can be annotated in PowerPoint or another image editor with the main results from our functional analysis, and any specific nodes of interest (Fig. 3a).
47. 3D module view. Click the '3D' icon in the vertical toolbar to view the network in 3D. Once the view finishes loading, click the highlighting icon (sun) to remove the seed node halos and change the background to white. Next, go to the 'Module Explorer' panel and perform module detection with the 'InfoMap' algorithm, following what was done in 2D. You should see the same module results as before. The modules may take ~20 s to finish loading—once they are done, every module will be highlighted with a colorful sphere in the network. Click the checkbox next to Modules 0–5 to highlight only these, and to reduce the opacity of the rest of the network. If you click and drag any node within one of the highlighted modules, the entire module will move. Perform some manual adjustments to make the 3D structure of all highlighted modules easier to see in one view. Then, click the edge bundling icon in the vertical toolbar. The two views—2D and 3D—offer complementary information. Individual nodes and their connections are easier to visualize in 2D, while the higher-level network connectivity between modules is easier to visualize in 3D, particularly when you can interactively rotate the view (Fig. 3b). Download this image using the same method as described in the previous step.
48. Context creation through signal enrichment. The final steps of Stage 2 will briefly demonstrate how OmicsNet can be used to add biological context to a smaller list of features by focusing on only the list of significant proteins. Return to the homepage (www.omicsnet.ca) and refresh to start a new session. Click 'Click here to start', select the 'Proteins' input, select Entrez ID, copy the list of 118 significant proteins, click 'Upload' and click 'Proceed'.

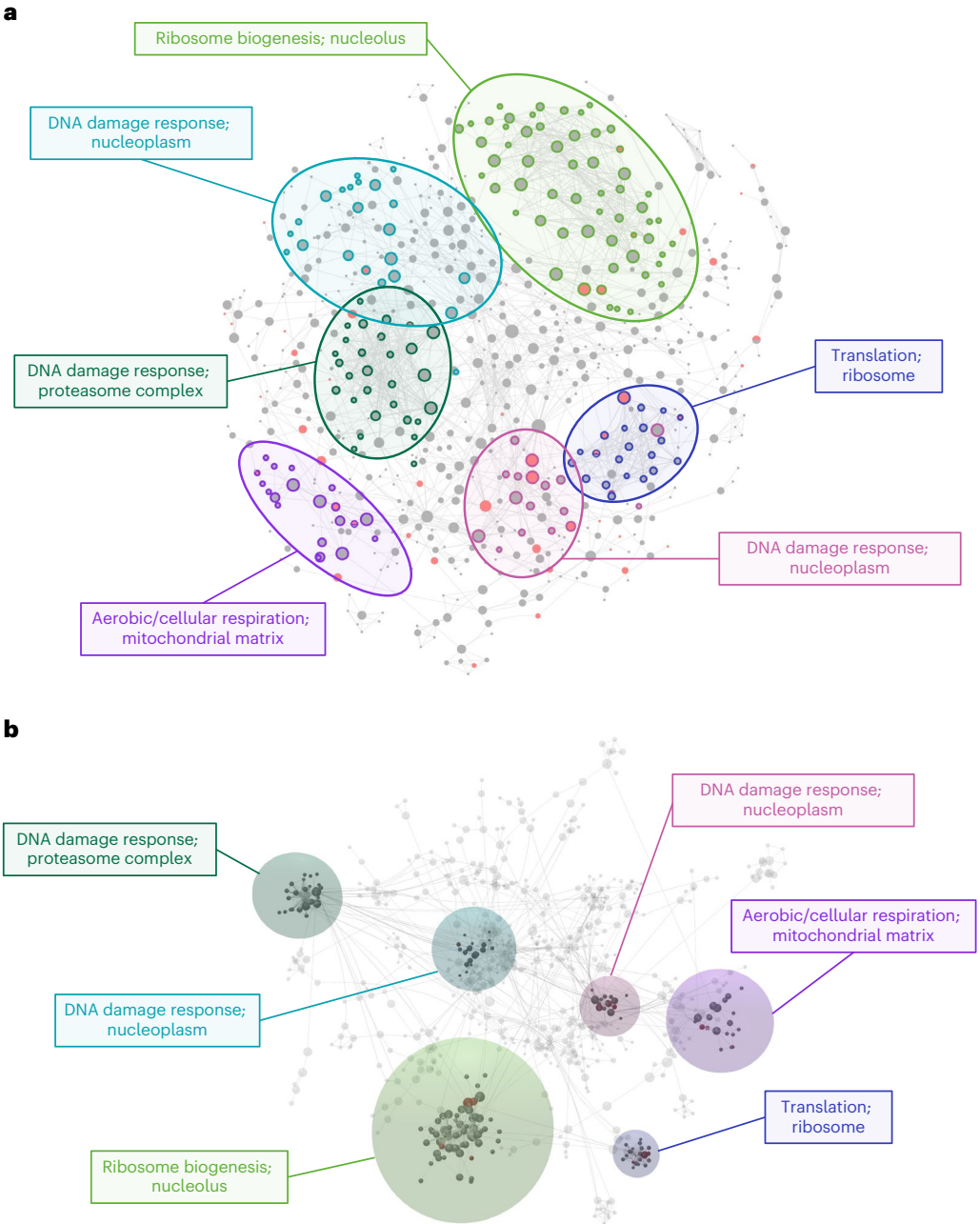


Fig. 3 | Knowledge-driven multi-omics network created through signal distillation with OmicsNet. a,b, Modules of densely connected features are highlighted and annotated with their most significant GO:BP (biological process) and GO:CC (cellular component) in 2D (**a**) and 3D (**b**). GO terms are slightly edited for length/clarity. For more module statistics, see Table 1.

49. Navigate to the 'Metabolite-protein' tab, select the 'Recon3D' database and click 'Submit', creating a network with size '230 – 311 – 35'. Recon3D describes over 13,000 metabolic reactions that incorporate both metabolites and proteins[76], enabling the retrieval of metabolites that are involved in metabolic reactions with the list of uploaded proteins. Next, navigate to the 'TF-gene' tab, select TRRUST[77], and click 'Submit'. This retrieves transcription factors that target genes that code for our list of proteins, in a network of size '92 – 86 – 18'. Click 'Proceed'.

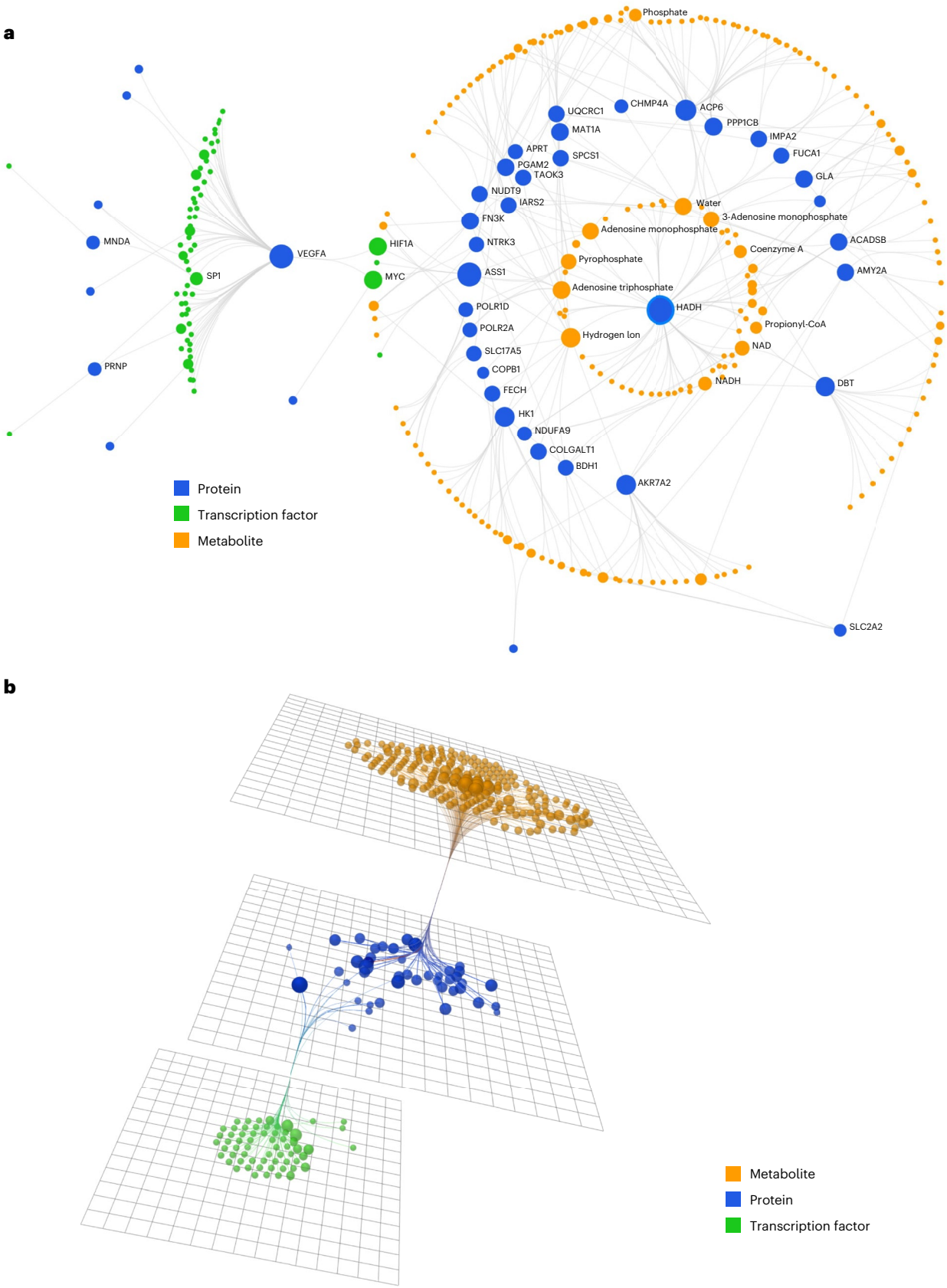

a
Phosphate
UQCRC1
CHMP4A
ACP6
MAT1A
PPP1CB
IMPA2
APRT
SPCS1
FUCA1
PGAM2
TAOK3
GLA
NUDT9
IARS2
Water
FN3K
3-Adenosine monophosphate
Adenosine monophosphate
MNDA
VEGFA
HIF1A
NTRK3
ACADSB
Coenzyme A
SP1
MYC
ASS1
Pyrophosphate
AMY2A
Adenosine triphosphate
POLR1D
HADH
POLR2A
Propionyl-CoA
Hydrogen Ion
SLC17A5
NAD
COPB1
NADH
DBT
PRNP
FECH
HK1
NDUFA9
COLGALT1
BDH1
AKR7A2
SLC2A2
Protein
Transcription factor
Metabolite
b
Metabolite
Protein
Transcription factor

Fig. 4 | Knowledge-driven multi-omics network created through signal enrichment with OmicsNet. a,**b**, Networks displaying the results of enriching significant proteins with interacting transcription factors and metabolites in 2D (**a**) and 3D (**b**).

50. On the network filtering page, subnetwork1 should have 43 seed nodes and 381 edges. Click 'Proceed' to go to the network viewer. All three node types are integrated together in a network, with metabolites in yellow, transcription factors in green and gene/proteins in salmon. Change the background to white and adjust the node colors until you are satisfied with the visual appearance. The 'Concentric Circle' layout can be used to organize the entire network around a feature of interest. The 'HADH' node seems to be central to the network—it has very high degree and betweenness statistics in the 'Node Explorer' table. It may also be relevant to human islet and diabetes research; loss of function mutations in HADH cause congenital hyperinsulinemia[78,79]. Double-click this node in the network and select 'Concentric Circle' from the layout dropdown. Make some manual adjustments to the nodes until all the labels are readable, then click the edge bundling icon (Fig. 4a).
51. 3D visualization. Go to the vertical toolbar inside the network viewer and click the '3D' icon. Toggle off seed node highlighting by clicking the icon that looks like a sun and change the background to white. Change the 'Layout' to 'Perspective' to arrange the nodes based on their layer and click the edge bundling icon in the vertical toolbar (Fig. 4b).

Stage 3: data-driven integration with OmicsAnalyst

● **TIMING ~45 min**

▲ **CRITICAL** The final stage will cover integrative analysis of omics data using joint DR (jDR) methods. This approach requires exactly matching samples across datasets. The lipidomics and transcriptomics data from Stages 1 and 2 overlap across 43 samples, while the proteomics and transcriptomics only overlap across 8 samples. As data-driven multi-omics integration requires large sample size, only the lipidomics and transcriptomics data will be integrated.

52. Starting up. Go to the OmicsAnalyst homepage (www.omicsanalyst.ca) and click 'Get Started'. By default, OmicsAnalyst assumes that omics data are already normalized using the optimal approach developed for the corresponding omics data type. We strongly recommend that users always thoroughly explore their data from a single-omics perspective first, and comprehensive options are available on ExpressAnalyst for transcriptomics/proteomics, MetaboAnalyst for metabolomics and MicrobiomeAnalyst for microbiome data.
53. Uploading data. Click 'Metadata table', select human for the species, choose the same metadata table that was uploaded to ExpressAnalyst and MetaboAnalyst, and click 'Submit'. Next, upload the normalized transcriptomics and lipidomics matrices that were saved during Stage 1. Click 'Omics Data #1', select 'Transcriptomics' for the data type, select 'Entrez ID' for the ID type, choose the 'normalized_rnaseq.csv' file that was saved earlier and click 'Submit'. Next, click 'Omics Data #2', select 'Metabolomics', leave the ID type as '– Not specified –', choose the 'normalized_lipids.csv' files and click 'Submit'. Click the 'Proceed' at the bottom.

    ▲ **CRITICAL STEP** Users can upload up to five omics datasets. Although there are three omics data in the original study, including the proteomics data would drastically decrease the sample size since only intersecting samples will be kept, making many DR methods unsuitable for data-driven integration. Please refer to Box 1 for more details on sample sizes.
54. Multi-omics data harmonization. Some downstream multi-omics integration methods are sensitive to imbalanced dimensionality, meaning that one omics layer will dominate contributions to the integrated results if it has many times more features or larger overall variations than the other layer(s) (Box 3). On this page, apply additional filtering and scaling to make the data more comparable across layers, as different omics data types often have very different number of features and scales of measurements, as follows:

BOX 3

Additional processing steps for multi-omics harmonization

Single-omics workflows generally include filtering steps to remove features that are either unreliable because they are near the detection limit, or uninteresting because they have low variance across all experimental groups. This improves the statistical power of downstream analyses. Normalization steps are employed to correct for systemic, technical factors (such as difference in volume or sequencing depth) across samples, and data are transformed to make their feature distributions more comparable and roughly normally distributed. For data-driven multi-omics integration, additional processing steps are often necessary to improve both the integration and visualization of the results.

Filtering

Some integration methods are sensitive to imbalanced dimensionality (i.e., one omics layer has many more features than the others). In this case, the omics type with more features will overshadow the other layer(s). For example, multi-omics factor analysis (MOFA) concatenates all features before computing components that maximally explain the variability in the dataset[19]. If one omics layer contributes several orders of magnitude more features, the results will be nearly the same as if the other omics types were not included at all. Pairwise correlation analysis also suffers from this problem. We can alleviate this problem by filtering the high-dimensional layer to include only the top features based on variability. This approach is described in Step 65 in Stage 3 of the procedure.

Scaling

Different omics layers usually have different value ranges that are characteristic of the underlying platforms, which makes integrated analysis and visualization challenging. Consider a heat map showing abundance values for multiple omics types, where each type has a distinct range of values. The only visible patterns will be the difference between omics types, whereas different values within a particular omics layer will be challenging to distinguish. Also, some DR methods compute components that maximize covariance within and across omics layers[20,45]. Since covariance is unscaled, the omics layer with a greater range of values will dominate the analysis. Scaling each omics layer to have comparable distributions can address these issues. This approach is described in Step 54 in Stage 3 of the procedure.

- Data filtering. Start without any additional filtering. It will be clear which methods are dominated by one omics data in the case of very imbalanced dimensionality. At this point, users can go back and apply filters as needed to address this problem. Leave the filters as the default (no filtering)
- Data scaling. Change 'Dataset' to 'Apply to all', leave the scaling method as 'Auto scaling' and click 'Update'. Notice that the overall distributions of transcriptomics and lipidomics data are now roughly the same in the density plot
- Metadata editing. Click the 'Metadata Overview' tab. The set of samples with overlapping transcriptomics and lipidomics data contain only four donors with ND. In the original publication, the authors relabeled all donors with IGT as ND to create a more balanced sample size across diagnosis groups. This is justified because single omics showed such a small difference between IGT and ND (recall that we had no significant features detected as differentially expressed in the transcriptomics analysis with ExpressAnalyst). Since there are only four ND donors, use the 'Edit' function to relabel them as 'IGT'. Click the 'Diagnosis' column header to sort by class, find each row with 'ND', click the pencil icon in the second last column, change 'ND' to 'IGT' and click the checkmark in the second last column. Click 'Edit metadata column', navigate to the 'Order (factor-level)' tab and re-order to 'IGT', 'T3D', 'T2D' and click 'Update'. Scroll to the bottom of the page and click 'Proceed'. Click 'Confirm' in the dialog warning on how missing values will be handled
  ▲ **CRITICAL STEP** Different omics data typically have different ranges of values. We recommend scaling features to make the range of values comparable across omics layers. Some integration methods identify key trends by maximizing measures of variability, and in this case, the omics layer with a greater range will dominate simply because its variability measures will be systematically higher. In addition, scaling can improve the appearance and visual interpretability of many multi-omics graphics.

55. Differential analysis. Select HbA1c as the primary metadata, include age and BMI as covariates, leave the blocking factor and *P* value inputs as is, and click 'Submit'. The same set of statistical relationships will be evaluated across all uploaded datasets and displayed in different tabs. There should be 46 lipids and 3,089 genes significantly associated

with HbA1c (adjusted *P* value <0.05). These are similar but slightly different from the single-omics results since there are slightly different sample numbers (only samples present in all omics datasets are included in the analysis). This differential analysis uses the same underlying R package as the differential analysis in ExpressAnalyst and MetaboAnalyst. Please see Steps 7–9 for a more in-depth explanation. Click 'Proceed'.

▲ **CRITICAL STEP** Do not worry if there are no statistically significant results from some of your datasets at this step. Integrative analyses are focused on understanding high-level patterns and relationships between sets of features (both within and across omics layers), and *P* values are only used as a means for feature prioritization or ranking.

56. Method selection. There are three tracks for data integration: (1) jDR, which computes major trends of variability across all omics datasets; (2) correlation network, for understanding pairwise relationships between key features; and (3) clustered heat map, for joint visualization of two omics datasets. Since each dataset was already analyzed in detail in Stage 1, this stage will focus on the DR and correlation network. Make sure that you are on the 'Dimensionality Reduction' tab.
57. jDR. Each of these methods computes components that explain the main trends of variability within the data. Compared with principal component analysis (PCA) commonly employed for single-omics DR, jDR methods compute components across multiple tables (i.e., individual omics data) at the same time. Their underlying algorithms maximize different criteria, producing slightly different results (Box 4). Select 'MCIA' (multiple co-inertia analysis) from the dropdown menu and click 'Proceed'. MCIA finds components

BOX 4

Understanding DR methods

There are many different DR methods in the literature. The most commonly used DR methods are probably PCA and partial least square–discriminant analysis (PLS–DA)—both are used primarily for single omics data analysis. PCA aims to identify a few new components that explain the most variance of the data, while PLS–DA aims to identify a few new components that explain the most covariance between data and phenotypes of interest. Joint dimensionality reduction (jDR) methods are variations of PCA or PLS–DA to simultaneously project multiple data tables to a low-dimensional space with or without consideration of class labels. Each method computes components that maximize some statistical terms. The maximized term can integrate multiple statistics, which is the key concept in jDR.

Symmetric versus asymmetric

DR methods can be based on either regression or correlation. Regression-based methods are asymmetric, because the order of the inputs matter. For example, $y - x$ does not equal $x - y$. Correlation-based methods are symmetric because cor(x, y) does equal cor(y, x). The same concept extends to multivariate methods, where the inputs are whole tables instead of single variables. Asymmetric methods have some assumptions, including that information is flowing in one direction (i.e., from *X* to *Y*), which may not always be true for omics data other than genomes (i.e., SNPs). For these reasons, we only include symmetric multivariate DR methods in OmicsAnalyst.

Shared versus complementary trends

In general, we cannot directly merge our omics matrices, even after scaling. This is because scaling does not address the 'shape' of the data distributions (also called the inertia, or mass distribution). General 'shape' patterns are usually distinct across omics layers, which can cause strange and unexpected results. Different methods address this problem in different ways. For instance, MCIA and DIABLO identify components separately but simultaneously in each layer, by maximizing a term that includes variance of each data and correlation across data. This finds a balance between components that both explain a substantial proportion of the variability within each layer and are shared across layers. MOFA first performs an additional normalization step to correct for systematic differences in 'shape'. Then, all omics features are directly merged into the same matrix, and subject to PCA. Since there is no stipulation that components should be correlated across layers, it is possible for some components to be almost 100% driven by one omics layer. This allows us to find both shared and complementary factors across omics layers.

Supervised versus unsupervised

This refers to whether class labels are considered while computing component. Both MCIA and MOFA are unsupervised as they only consider omics data, while DIABLO is supervised as it also considers the variance of a single metadata variable. DIABLO has a covariance parameter (between 0 and 1) that determines how much the covariance between omics layers is weighted relative to the variance of the specified metadata. Lower values of the parameter make the 'supervised' component more influential. A value of 1 will maximize covariance across omics layers (i.e., does not consider the metadata at all), making it very similar to MCIA.

that simultaneously maximize sources of variability within each dataset, and correlation of the components across datasets. This means that MCIA components capture variability trends that are shared across all omics datasets.

58. Overview of jDR result summary. View the summary plots to see how the top components relate to the individual omics data and to the metadata labels. The plot on the left shows pairwise scores plots for the first five components colored based on the 'Diagnosis' groups. Components 1–3 showed partial separation while components 4 and 5 showed almost complete overlap of the three diagnosis groups (Fig. 5a). The second plot shows the percentage of the variability within each omics dataset that is explained by each component. A detailed table with corresponding loadings is available for exploration—click the table icon in the 'Browse' column at the bottom of the page to view the loadings for each dataset.
▲ **CRITICAL STEP** MCIA is an unsupervised method and does not consider the 'Diagnosis' labels when computing the top components. The component score plot summarizes shared patterns of variation in the data. These trends could be from biological/environmental (i.e., sex, age, BMI, diagnosis) or technical (i.e., batch, sample preparation parameters) sources. Annotating the computed component scores with metadata helps interpret the main trends in our data.
59. Visualizing the MCIA result. Click 'Proceed' to visualize the main jDR results in a 3D scatterplot. By default, a scores plot of the top three components is displayed, with points colored by the 'Diagnosis' metadata and overlain with ellipses showing the 50% confidence region for each 'Diagnosis' class. The confidence level can be adjusted, for example, to 95%, by clicking the 'Advanced Options' button in the top horizontal bar and adjusting the 'Confidence' slider on the first tab. The 3D scatterplot can be rotated by clicking and dragging anywhere within the plot viewer. The 'T3D' group completely overlaps with the 'T2D' and 'IGT' groups and parts of the 'T2D' and 'IGT' groups are separated. It is also possible to color the sample nodes using different metadata under the 'Overview' panel. For instance, selecting 'HbA1c' will color samples using a red color gradient.
60. By default, the 'Ranked Features' panel shows the significant features identified from the differential expression analysis that was performed in Step 55 to identify features associated with HbA1c. Make sure the 'Omics type' is 'Transcriptomics' and click the top three rows (PCOLCE2, HSD17B13 and ALDOB) to project the feature loadings onto the scores plot. This is essentially a simplified biplot (see the subsection 'Using DR for data-driven multi-omics integration'). These three features all point in the direction of T2D samples, meaning they have higher abundance in donors with T2D. This makes sense based on previous studies, which have observed that ALDOB transcripts are positively correlated with HbA1c, meaning that they have higher expression in donors with T2D[80]. Next, change the 'Omics' dropdown to 'Metabolomics' and click the top three rows here (TAG.53.4.0, TAG.51.3.0 and PC.O.16.1.0.18.1.0). Now, the top features from both omics types are present in the same plot, showing how both layers contribute towards the observed separation of the 'Diagnosis' classes. Click the reset icon (counterclockwise circular arrow) in the vertical tool bar to reset the view and double-click the sample group icon (ellipse with points inside) to add back the 'Diagnosis' class confidence regions.
▲ **CRITICAL STEP** Intuitively, you can think about the feature loadings 'pushing' the sample scores in the directions that their arrows point, as these features show increased expression or abundance levels in the samples that the arrows point toward. If an arrow points in the direction of samples annotated with 'T2D', it can be interpreted that this feature has higher levels in donors with 'T2D'.
61. Identifying and exploring important pathways. Inside the 'Ranked Features' panel, change the 'Omics type' dropdown to 'Transcriptomics' and click 'Update'. Now, go to the panel directly below called 'Enrichment Analysis', leave the defaults and click 'Submit'. This will perform ORA on the list of significant features against the KEGG pathways. Clicking a pathway row will project loadings from the significant features in that pathway onto the component plot, enabling visualization of how that pathway contributes toward the observed sample separation patterns. For example, click the 'Citrate Cycle (TCA Cycle)' pathway—all pathway hits are pointing in the same direction, toward the 'IGT' samples,

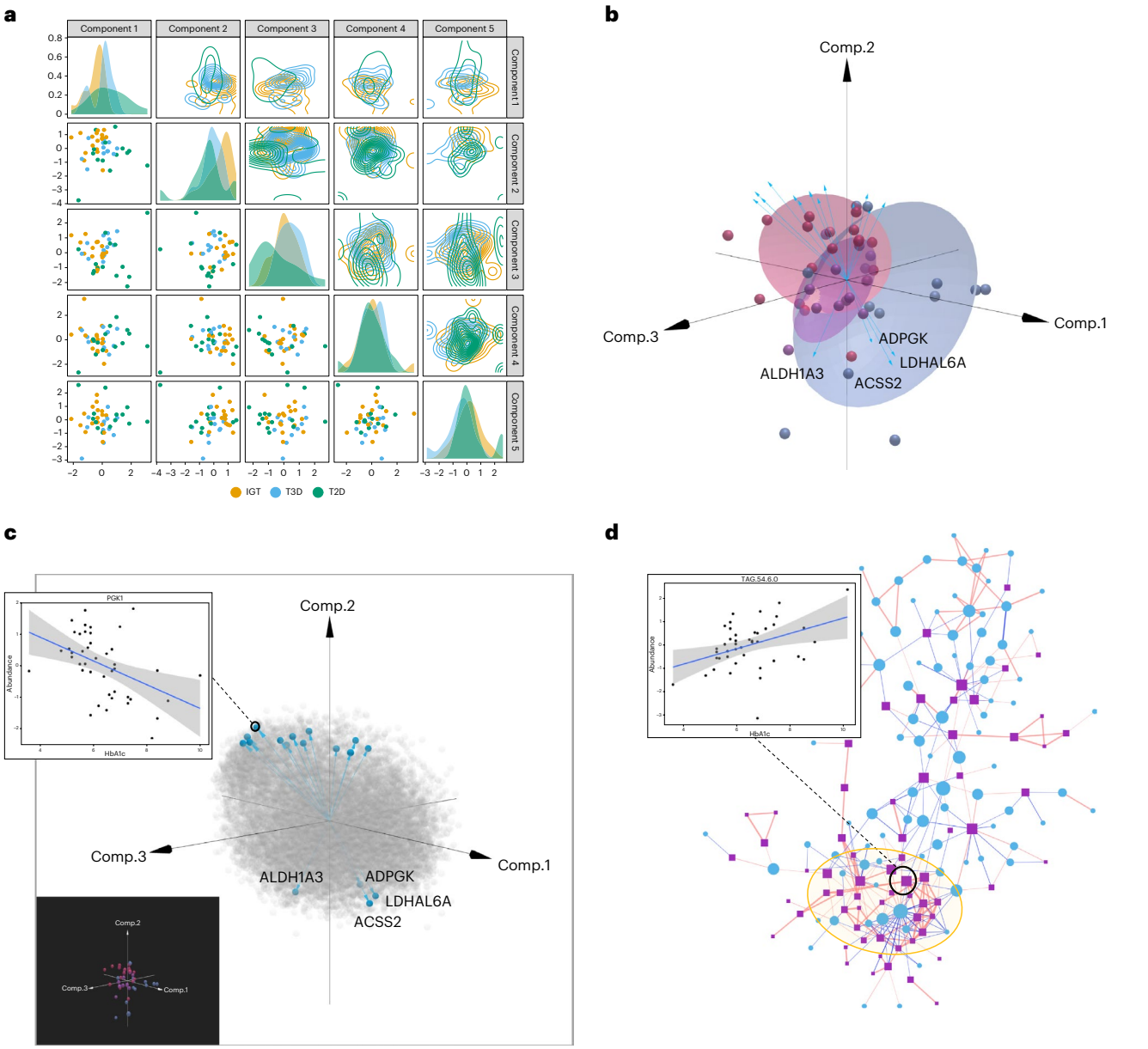


Fig. 5 | Data-driven MCIA results from OmicsAnalyst. a, Pairwise component scores plot for the top five components, annotated with the 'Diagnosis' metadata (IGT in orange, T3D in blue, T2D in green). **b**, 3D scatterplot of the top three components with four high-magnitude component 2 transcriptomic feature loadings from the 'Gycolysis/Gluconeogenesis' KEGG pathway projected onto the plot as vectors (labeled ADPGK, LDHAL6A, ACSS2 and ALDH1A3). **c**, Loadings plot with the same highlighted transcriptomic features from **b** in the background scatterplot, with the component scores scatterplot in the bottom inset with the black background, and a scatterplot of the individual *PGK1* mRNA levels (*y* axis) and associated donor HbA1c values (*x* axis) in the top inset with the white background, which is generated by clicking the PGK1 loading vector. **d**, Correlation network of high-magnitude loadings from the top three components. The inset in **d** is a scatterplot of TAG.54.6.0 lipid levels and associated donor HbA1c values, generated by clicking the TAG.54.6.0 node in the network. In the PGK1 and TAG.54.6.0 scatterplots, the gray area shows the 95% confidence interval of the fitted regression model. Feature names in **a**–**d** were manually added to the figure. Within OmicsAnalyst, users can access this information by hovering their mouse over the feature vectors.

which means they have higher expression values in these donors. Click the 'Citrate Cycle (TCA Cycle)' pathway row again to remove the loadings from the plot.

▲ **CRITICAL STEP** If you are having trouble viewing the pathway names, hover your mouse over the left boundary of the 'Ranked Features' and 'Enrichment Analysis' side panel. An arrow pointing left should appear, at which point you can click and drag to expand the size.

You can also change the amount of space for each column by hovering your mouse over the lines separating the columns in the table header, clicking and dragging. This is why we suggest using a minimum 15 inch screen and a mouse for a better user experience.

62. Understanding loadings and functions. The loadings for each component indicate the relative contribution of each feature. To display the top loadings for each component, go to the 'Ranked Features' panel, change the 'View' from 'Significant' to 'Component 1' and click 'Update'. This will show the feature loading scores, sorted by their absolute values. Click the top transcript (CCDC51) and note that the loading arrow points in mainly the same direction as the Component 1 axis. Click the 'CCDC51' row again to remove the loading from the component scores plot.
63. Create a list of features that can be analyzed by either ORA or GSEA to understand the biological processes underlying the variability captured by each component. For ORA, the list of features used for the analysis is the top 10% with the highest magnitude loadings. Go to the 'Enrichment Analysis' panel and perform ORA for the 'KEGG (gene)', 'Reactome' and 'GO:BP' databases. The biological themes of the significant results for Component 1 are mainly related to transcription. Next, change the 'Ranked Features' to Component 2 (still transcriptomics) and click 'Update'. Perform ORA with the 'KEGG (gene)' database. Here, the sixth most significant pathway is 'Glycolysis/Gluconeogenesis', which is highly relevant to diabetes. Click this pathway to project the pathway features onto the component scores plot. Four features (ALDH1A3, ACSS2, LDHAL6A and ADPGK) are pointing in the direction of the T2D samples, while the others are pointing in the direction of the IGT samples. ALDH1A3 was recently identified as a marker gene for beta cell dedifferentiation, a process that may underly beta cell dysfunction in patients with T2D[81] (Fig. 5b).
64. Switch to the loadings plot by clicking the switch views icon (two arrows pointing to each other) in the vertical toolbar. Now, the viewer shows all the feature loadings instead of the sample component scores. Change the 'Biplot Arrow' dropdown to 'No Label' in the top horizontal toolbar and select the 'Glycolysis/Gluconeogenesis' pathway to highlight the same features as before. Double-click one of the highlighted features to view its association with any metadata of interest (Fig. 5c).
65. *MOFA*. Click the 'Method Selection' link on the navigation bar at the top of the page. Go to the DR panel, select 'MOFA' (multi-omics factor analysis) and click 'Proceed'. In contrast to MCIA, MOFA performs a single PCA after normalizing for mass distribution (also termed 'inertia') and merging all features across omics layers. This means that components with correlated trends across omics layers are not explicitly prioritized, and top components can include trends that are only present in one or some of the omics layers. This can be an advantage as MOFA can retrieve both shared and complementary trends across omics layers. The plots on the 'Overview' tab show that most components, and especially Component 1, are much more relevant to the transcriptomics data. This is common in MOFA results when there is a substantial imbalance in the number of omics features in each layer. Recall that there are 16,342 transcripts and only 116 lipids in the final processed datasets. One option is to navigate back to the 'Quality Checking' page and apply additional filters to the RNA-seq data to make the dimensionality more similar, however in this case the number of features is so different that a better strategy is to use methods such as MCIA and DIABLO that are inherently robust to imbalanced dimensionality. Click 'Previous'.
66. *DIABLO*. Go to the DR panel, select DIABLO and click 'Proceed'. DIABLO is like MCIA because it prioritizes trends that are shared across layers. However, it also maximizes a term that describes the separation of categorical metadata groups and is therefore a supervised method. Users can adjust the 'covariance parameter' to balance between the ability of the components to capture the covariance across omics data, or to explain metadata of interest (i.e., group separation). By default, DIABLO is conducted using the primary metadata and a covariance parameter of 0.2 (Box 4). The 'Dimensionality Reduction Overview' tab shows that the components are fairly balanced across the lipid and transcriptomics data, and the 'Sample Score Plot' tab shows that the 'Diagnosis' classes are more separated compared with MCIA, which is expected since DIABLO is a supervised method. Click 'Proceed'.

67. Click the scatterplot and drag to rotate the view. The greater separation between Diagnosis classes is more evident in 3D. Perform GSEA on the top three transcriptomics components with the KEGG, Reactome and GO databases. Most of the top significant results overlap with the results from MCIA. When multiple statistical methods highlight the same biological themes, it increases confidence that these patterns are robust and biologically meaningful. Keep in mind that while functional analysis was only performed on the transcriptomics data, the transcript loading scores were prioritized based on their high covariance with the lipidomics data. Use the top navigation bar to go back to the 'Method Selection' page.
68. Performing correlation network analysis. This step will use pairwise correlation calculations coupled with network visualization to understand relationships between the top features in the top components constructed from the DR methods. Click the 'Correlation Network' tab, select 'MCIA' for the 'Feature selection method' and click 'Proceed'. By default, OmicsAnalyst will take the top 20 features with the highest absolute loadings from each of the top three components and calculate the pairwise correlation between omics layers. Any correlation with absolute values greater than 0.5 (default) is retained as a feature-feature connection, and then all connections are merged into interconnected subnetworks (see the 'Network summary' tab at the bottom of the page).
69. Refining correlation networks. By default, OmicsAnalyst computes pairwise correlations only between omics types. Uncheck the 'Between-omics only' checkbox to tell OmicsAnalyst to calculate pairwise correlations both between and within omics layers, so relationships between pairs of lipids and pairs of transcripts will be calculated in addition to the lipid–transcript pairs described earlier. There are often systematic differences in feature distribution range and shape across omics layers due to differences in platform technology and data processing steps, which reduces the statistical power for calculating correlations between features of different omics types[9,59]. Since between-omics correlations are usually systematically lower than within-omics correlations, OmicsAnalyst has two separate thresholds to allow users to prioritize the top connections in each category. View the 'Correlation Summary' plot at the bottom of the page to see the distribution of between-omics and within-omics correlations. Adjust some of the other filters to increase the number of features in the analysis. Decrease the 'Within-omics' filter from 0.9 to 0.8 and increase the 'Top feature number per component' from 20 to 50. Click 'Update'. The largest 'subnetwork1' should contain 94 transcripts, 71 lipids and 382 edges. Click 'Proceed' to view 'subnetwork1'.

    ◆ **TROUBLESHOOTING**
70. Visualizing correlation networks. The largest network (subnetwork1) is displayed in the network viewer. Note that the layout is random, and so while you should have the same features and edges, their positions will probably be different than the network shown in Fig. 5d. In the network, the blue circles represent genes, the purple squares represent lipids, and edges between features indicate that the pairwise correlations between the two connected features passed the thresholds set in the previous page. The thickness of each edge represents the strength of the correlation and the color represents the direction, with red for positive correlations and blue for negative correlations. The size of each feature corresponds to the betweenness of each node. Nodes with high betweenness centrality values are potentially more important as they fall on many of the shortest paths between pairs of nodes in the network. Hover your mouse over an edge and click to view details in the 'Current Selections' panel on the lower lefthand side. This shows the two omics feature nodes that this edge connects, as well as the correlation coefficient for these two omics features.
71. OmicsAnalyst has several tools for adjusting the network aesthetics. Select 'Label' from the 'Styling' dropdown, go to the 'Display' tab, change the option to 'Hide' and click 'Submit'. Change the 'Layout' dropdown to 'Large Graph Layout'. Change the 'Layout' dropdown again to 'Reduce Overlap'. The main group of densely connected nodes are all TAG lipids (triacylglycerols lipid class) that are positively correlated with each other and negatively correlated with a few genes, including CCDC51 and NUP54 (Fig. 5d). Double clicking a feature will show its association with HbA1c. For example, clicking 'TAG.54.6.0' shows that

it is positively associated with HbA1c. Module detection and functional analysis could be performed on this network using the methods described in Stage 2.

▲ **CRITICAL STEP** This type of analysis is well suited for functional interpretation of omics features with limited annotation. If a feature with unknown function is strongly connected to many features that all participate in a distinct biological process, it is likely that the unknown feature participates in this process as well. This is known as 'guilt-by-association' functional annotation[82]. In the case of these data, it may be helpful use the transcriptomics results to understand biological processes that could be related to the lipid abundance levels. Note that the interpretation is less straightforward for this data since the lipids and genes were measured in different tissues, but it serves to illustrate the general concept.

Troubleshooting

Troubleshooting advice can be found in Table 2.

Table 2 | Troubleshooting table

Step	Problem	Possible reason	Solution
4	Data uploading failed	The files are not formatted correctly	See our extensive formatting details on each tool, under the 'Data Formatting' tab
8 (and others)	The number of significant features does not match those described in the procedure	Updates have been made to the statistical pipeline, databases or default parameters	Check the recent updates to see if any obvious changes were made, for example, an update in the gene annotation files. If so, exact numbers may vary slightly throughout the protocol, but the main trends and patterns should stay the same
11	The diagnosis groups in the violin plot do not match Fig. 2a	You did not rearrange the metadata order	Go to the 'Metadata Overview' tab on the 'Data Quality Check' page and click 'Edit metadata column'. Use the 'Order' tab to set the order
35	The final network size does not match those described in the procedure	All database options reset to default when you select a new list. You did not select the correct database the second time	Make sure to select the correct database in Step 35 (STRING, protein–protein tab)
39	The network layout does not exactly match the network in Fig. 3a	Most network layouts are stochastic and change slightly each time	Verify that the number of nodes and edges are the same as reported in Step 36 of the procedure, and that the layout is just in a different orientation
69	The number of network nodes and edges is different than described in Step 69 of the procedure	Some parameter values were reset when you unselected 'Between-omics only'	Selecting/unselecting resets some backend parameters within OmicsAnalyst. Make sure the parameters exactly match those specified in Step 69 of the procedure before you click 'Update'

Timing

The complete protocol can be executed in ~2 h, with Stages 1–3 expected to take 45 min (Steps 1–31), 30 min (Steps 32–51) and 45 min (Steps 52–71), respectively.

Anticipated results

In this protocol, we show results for one multi-omics dataset, analyzed with four different web-based tools (ExpressAnalyst, MetaboAnalyst, OmicsNet and OmicsAnalyst). The main graphical outputs are shown in Figs. 2–5. Other results are summarized separately below.

Stage 1: single-omics workflows

Each single-omics workflow produces multiple versions of the data after the various processing and analytical steps. During the protocol, we explicitly download the annotated, filtered

and normalized omics tables for the lipidomics and transcriptomics data, and tables of the differential expression results for the proteomics and transcriptomics data. The normalized transcriptomics data should have 16,353 features for 133 donors. The normalized lipidomics data should have 116 features for 43 donors. There should be 1,678 and 118 significant features from the T2D–ND differential analysis for the transcriptomics and proteomics data, respectively.

Stage 2: knowledge-driven integration with OmicsNet

The knowledge-based integration produces two main networks, one using the 'signal distillation' approach, and the other using the 'signal enrichment' approach. The final network for 'signal distillation' (Steps 32–50) should have 667 seed nodes with 1,822 edges. The final network for 'signal enrichment' (Steps 48–51) should have 43 seed nodes with 381 edges.

Stage 3: data-driven integration with OmicsAnalyst

During data-driven integration, we produce 3D scatterplots for joint dimensionality reduction with the MCIA, MOFA and DIABLO methods, and a correlation network with top features from the MCIA analysis. The final correlation network should have 94 genes, 71 lipids and 382 edges.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All datasets used in this protocol are from a previously published study that collected multi-omics data from islet tissue (transcriptomics and proteomics) and plasma (lipidomics) from human donors undergoing pancreatic surgery[30]. The example datasets used in this protocol are integrated as examples throughout the four web tools. The unprocessed matrices are included as examples in ExpressAnalyst (www.expressanalyst.ca) and MetaboAnalyst (www.metaboanalyst.ca). Lists of significant features are included as examples in OmicsNet (www.omicsnet.ca). Normalized matrices are included as examples in OmicsAnalyst (www.omicsanalyst.ca). See the 'Materials' section for further information about the example datasets.

Code availability

ExpressAnalyst, MetaboAnalyst, OmicsNet and OmicsAnalyst are all freely available as web-based applications. The underlying R packages for each tool are freely available as GitHub repositories: ExpressAnalystR (https://github.com/xia-lab/ExpressAnalystR), MetaboAnalystR (https://github.com/xia-lab/MetaboAnalystR), OmicsNetR (https://github.com/xia-lab/OmicsNetR) and OmicsAnalystR (https://github.com/xia-lab/OmicsAnalystR) under the GNU General Public License version 2 or later.

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Author contributions

J.D.E. and J.X. prepared the manuscript. J.D.E., G.Z., Y.L. and J.X. contributed to the development of the tools (MetaboAnalyst, ExpressAnalyst, OmicsNet and OmicsAnalyst). J.K., C.E., J.D.J. and P.E.M. validated the tools and protocol steps, resulting in improvements to both based on their feedback. All authors read and approved the final manuscript.

Competing interests

J.D.E., G.Z. and J.X. own shares of OmicSquare Analytics Inc.

Additional information

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Correspondence and requests for materials should be addressed to Jianguo Xia.

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Our web collection on statistics for biologists contains articles on many of the points above.

Software and code

Policy information about availability of computer code

Data collection	No data was collected for this protocol. All data were previously published, and the original publication describing the data is noted in the text.
Data analysis	This manuscript is a protocol for how to analyze multi-omics datasets with software that we created. All software is open-source, and we provided Github repository URLs in the code availability statement at the end of the manuscript.

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All datasets used in this protocol are from a previously published study that collected multi-omics data from islet tissue (transcriptomics and proteomics) and plasma (lipidomics) from human donors undergoing pancreatic surgery30. The example datasets used in this protocol are integrated as examples throughout the four web-tools. The unprocessed matrices are included as examples in ExpressAnalyst (www.expressanalyst.ca) and MetaboAnalyst (www.metaboanalyst.ca). Lists

of significant features are included as examples in OmicsNet (www.omicsnet.ca). Normalized matrices are included as examples in OmicsAnalyst (www.omicsanalyst.ca). See the Materials section for further information about the example datasets.

Human research participants

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Reporting on sex and gender	Metadata describing sex was not provided by the authors who originally published the data re-analyzed in this protocol manuscript, for what we assume are privacy reasons as identification with sex alongside the other provided metadata is highly possible. However, our manuscript clearly describes how sex can be statistically accounted for during analysis (step 8).
Population characteristics	This protocol re-analyzed previously published data. All metadata made publicly available by the original authors is provided to researchers executing the protocol (diabetes diagnosis, age, oral glucose tolerance test, HbA1c levels, BMI).
Recruitment	No new data was collected in this work, therefore no recruitment was performed.
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